

Quantifying the shape of cells - from Minkowski tensors to p -atic order

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This paper introduces an **important** theoretical method for characterizing symmetries of cells in biological tissues by capturing their real shape, and it sets results in contrast to related methods. The robustness of the paper's method to correctly capture dynamic and geometric changes that the cells may undergo is determined by **convincing** computational models, but the experimental support is **incomplete** and would benefit from better experimental imaging with higher quality and extended analysis. This would not only support the advantage of this method, but also strengthen its application to biological systems.

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Abstract

P -atic liquid crystal theories offer new perspectives on how cells self-organize and respond to mechanical cues. Understanding and quantifying the underlying orientational orders is therefore essential for unraveling the physical mechanisms that govern tissue dynamics. Due to the deformability of cells this requires quantifying their shape. We introduce rigorous mathematical tools and a reliable framework for such shape analysis. Applying this to segmented cells in MDCK monolayers and computational approaches for active vertex models and multiphase field models challenges previous findings and opens new pathways for understanding the role of orientational symmetries and p -atic liquid crystal theories in tissue mechanics and development.

Introduction

The importance of orientational order in biological systems is becoming increasingly clear, as it plays a critical role in processes such as tissue morphogenesis, collective cell motion, and cellular extrusion. Orientational order results from the shapes of cells and their alignments with

neighboring cells. Disruptions in this order, known as topological defects, are often linked to key biological events. For instance, defects—points or lines where the order breaks down—can drive cell extrusion (Saw et al., 2017 [DOI](#); Monfared et al., 2023 [DOI](#)) or trigger morphological changes in tissues (Maroudas-Sacks et al., 2021 [DOI](#); Ravichandran et al., 2024 [DOI](#)). Orientational order is linked to liquid crystal theories and should here be interpreted in a broad sense. Recent evidence extends beyond nematic order, characterized by symmetry under $180^\circ = 2\pi/2$ rotation, to higher-order symmetries such as tetratic order ($90^\circ = 2\pi/4$), hexatic order ($60^\circ = 2\pi/6$), and even general p -atic orders ($2\pi/p$, p being an integer). In biological contexts, nematic order ($p = 2$) has been widely studied in epithelial tissues (Duclos et al., 2017 [DOI](#); Saw et al., 2017 [DOI](#); Kawaguchi et al., 2017 [DOI](#)), linking defects to cellular behaviors and tissue organization. More complex orders, such as tetratic order ($p = 4$) (Cislo et al., 2023 [DOI](#)) and hexatic order ($p = 6$) (Li and Ciamarra, 2018 [DOI](#); Durand and Heu, 2019 [DOI](#); Pasupalak et al., 2020 [DOI](#); Armengol-Collado et al., 2023 [DOI](#); Eckert et al., 2023 [DOI](#)), have also been observed in experimental systems, offering new perspectives on how cells self-organize and respond to mechanical cues. Understanding and quantifying orientational order and the corresponding liquid crystal theory are therefore essential for unraveling the physical mechanisms that govern biological dynamics.

In physics, p -atic liquid crystals illustrate how particle shape and symmetry influence phase behavior, with seminal works dating back to Onsager's theories (Onsager, 1949 [DOI](#)). Most prominently, hexatic order ($p = 6$) has been postulated and found by experiments and simulations as an intermediate state between crystalline solid and isotropic liquid in (Halperin and Nelson, 1978 [DOI](#); Nelson and Halperin, 1979 [DOI](#); Murray and Van Winkle, 1987 [DOI](#); Bladon and Frenkel, 1995 [DOI](#); Zahn et al., 1999 [DOI](#); Gasser et al., 2010 [DOI](#); Bernard and Krauth, 2011 [DOI](#)). Other examples are colloidal systems for triadic platelets (Bowick et al., 2017 [DOI](#)) or cubes (Wojciechowski and Frenkel, 2004 [DOI](#)), which lead to p -atic order with $p = 3$ and $p = 4$, respectively. Even pentatic ($p = 5$) and heptatic ($p = 7$) liquid crystals have been engineered (Wang and Mason, 2018 [DOI](#); Yu and Mason, 2023 [DOI](#)). The corresponding liquid crystal theories for p -atic order have only recently been defined (Giomi et al., 2022 [DOI](#); Krommydas et al., 2023 [DOI](#)) and can also be used in biological contexts. However, while in colloidal systems particle shapes remain fixed, in biological tissues, cells are dynamic: their shapes are irregular, variable, and influenced by internal and external forces. These unique properties make quantifying p -atic order in tissues significantly more challenging, as quantification of the cell shapes is required. We will demonstrate that existing methods, such as bond-orientational order (Nelson and Halperin, 1979 [DOI](#)) or polygonal shape analysis (Armengol-Collado et al., 2023 [DOI](#)), often fail to capture the nuances of irregular cell shapes, which has severe consequences on the definition of p -atic order.

To address this, we adapt Minkowski tensors (Mecke, 2000 [DOI](#); Schröder-Turk et al., 2013 [DOI](#))—rigorous mathematical tools for shape analysis—to quantify p -atic order in cell monolayers. Minkowski tensors provide robust and sensitive measures of shape anisotropy and orientation, accommodating both smooth and polygonal shapes while remaining resilient to small perturbations. By applying these tools, we re-examine previous conclusions about p -atic order in epithelial tissues and demonstrate that certain widely accepted results require reconsideration.

Our study leverages a combination of experimental and computational approaches. Experimentally, we analyze confluent monolayers of MDCK (Madin-Darby Canine Kidney) cells, comparing wild-type tissues with E-cadherin knockout variants to explore the role of cell-cell adhesion in orientational order. **Figure 1** [DOI](#) provide a snapshot of the considered monolayer of wild-type MDCK cells, together with their shape classification by Minkowski tensors. The corresponding statistical data and probability distributions of these quantities are shown in **Figure 2** [DOI](#). These data already indicate no clear preference of a specific p -atic order and the necessity for robust and sensitive shape classification. Computationally, we employ two complementary models for cell monolayers: the active vertex model and the multiphase field

model. These approaches allow us to systematically vary parameters such as cell activity and mechanical properties, providing a comprehensive view of how p -atic order emerges in different contexts.

By combining these robust mathematical tools, experimental insights, and computational models, we establish a reliable framework for quantifying p -atic order in biological tissues. This framework not only challenges previous findings but also opens new pathways for understanding the role of orientational symmetries in tissue mechanics and development.

Methods and materials

Minkowski tensors

Essential properties of the geometry of a two-dimensional object are summarized by scalar-valued size measures, so called Minkowski functionals. They are defined by

$$W_0(C) = \int_C dC, \quad W_1(C) = \int_{\partial C} d\partial C, \quad \text{and} \quad W_2(C) = \int_{\partial C} \mathcal{H} d\partial C, \quad (1)$$

where C is the smooth two-dimensional object, with contour ∂C and $\mathcal{H}$ denotes the curvature of the contour ∂C (see [Figure 3](#) for a schematic description), as reviewed in [Mecke \(2000\)](#). They describe (up to field-specific prefactors) the area, the perimeter and a curvature-weighted integral of the contour, respectively. Minkowski functionals are also known as intrinsic volumes. For convex objects, they have been shown to be continuous and invariant to translations and rotations. Due to these properties and Hadwiger's characterization theorem ([Hadwiger, 1975](#)), they are natural size descriptors that provide essential and complete information about invariant geometric features. However, these properties also set limits to their use as shape descriptors. Due to rotation invariance, they are unable to capture the orientation of the shape. To describe more complex shape information the scalar-valued Minkowski functionals are extended to a set of tensor-valued descriptors, known as Minkowski tensors. These objects have been investigated both in mathematical ([Alesker, 1999](#); [Hug et al., 2008](#); [2007](#); [McMullen, 1997](#)) and in physical literature ([Schröder-Turk et al., 2010a](#); [Kapfer et al., 2012](#)). Using tensor products of the position vectors $\mathbf{r}$ and the normal vectors $\mathbf{n}$ of the contour ∂C , defined as $\mathbf{r}_a \odot \mathbf{n}_b = \mathbf{r} \odot \dots \odot \mathbf{r} \odot \mathbf{n} \odot \dots \odot \mathbf{n}$, the first considered a times and the last b times with $\odot$ denoting the symmetrized tensor product. The Minkowski tensors are defined as

$$\mathbf{W}_0^{a,0}(C) = \int_C \mathbf{r}^a dC, \quad \mathbf{W}_1^{a,b}(C) = \int_{\partial C} \mathbf{r}^a \odot \mathbf{n}^b d\partial C, \quad \text{and} \quad \mathbf{W}_2^{a,b}(C) = \int_{\partial C} \mathbf{r}^a \odot \mathbf{n}^b \mathcal{H} d\partial C. \quad (2)$$

For $\mathbf{W}_j^{0,0}(C) = W_j(C)$ for $j = 0, 1, 2$, so the Minkowski tensors are extensions of the Minkowski functionals. In analogy to Minkowski functionals also Minkowski tensors have been shown to be continuous for convex objects. Also, analogies to Hadwiger's characterization theorem exist ([Alesker, 1999](#)). However, for our purpose, the essential property is the continuity. It makes Minkowski functionals and Minkowski tensors a robust measure as small shape changes lead to small changes in the shape descriptor. Due to this property, Minkowski tensors have been successfully used as shape descriptors in different fields, e.g. in materials science as a robust measure of anisotropy in porous ([Schröder-Turk et al., 2010b](#)) and granular material ([Schröder-Turk et al., 2013](#)), in astrophysics to describe morphology of galaxies ([Beisbart et al., 2002](#)), and in biology to distinguish shapes of different types of neuronal cell networks ([Beisbart et al., 2006](#)) and to determine the direction of elongation in multiphase field models for epithelial tissue ([Mueller et al., 2019](#); [Wenzel and Voigt, 2021](#); [Happel and Voigt, 2024](#)). However, the mentioned examples exclusively use lower-rank Minkowski tensors $a + b \leq 2$. Higher rank tensors have not been considered in such applications but the theory also guarantees robust description of

Figure 1.

Shape classification of cells in wild-type MDCK cell monolayer. a) Raw experimental data. b) - f) Minkowski tensor, visualized using ϑ_p and q_p , **Equation 7** (see Methods and materials) for $p = 2, 3, 4, 5, 6$, respectively. The brightness and the rotation of the p -atic director indicates the magnitude and the orientation, respectively.

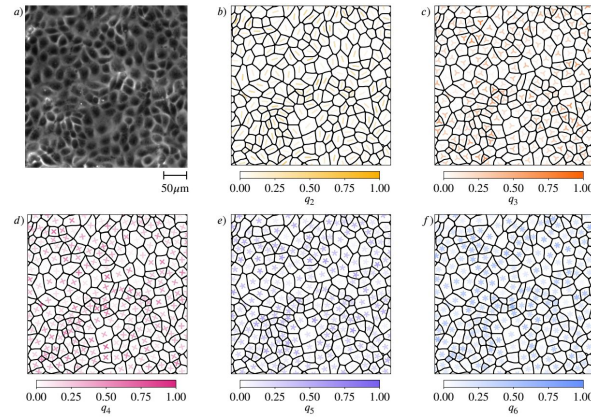


Figure 2.

Statistical data for cell shapes identified in **Figure 1** (see Methods and materials). a) Mean q_p and standard deviation $\sigma(q_p)$ of q_p . b) - f) Probability distribution function (PDF) of q_p for $p = 2, 3, 4, 5, 6$, respectively. Normal distributions with the same mean and standard deviations are added to the histograms as a guide to the eye.

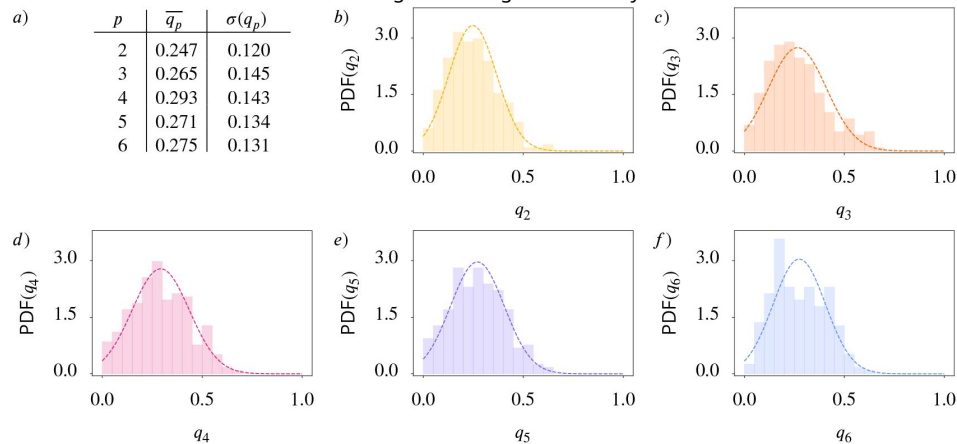
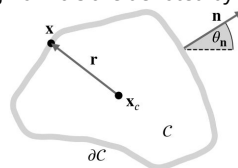


Figure 3.

Schematic description of a two-dimensional object with contour ∂C . We denote the center of mass with $\mathbf{x}_c$ and vectors from $\mathbf{x}_c$ to points $\mathbf{x}$ on ∂C with $\mathbf{r}$. The outward-pointing normals are denoted by $\mathbf{n}$, the corresponding angle with the x-axis by θ_n .



p -atic orientation for $p = 3, 4, 5, 6, \dots$. These results hold for smooth contours but also can be extended to polyconvex shapes. Furthermore, known counter-examples for non-convex shapes have very little to no relevance in applications (Kapfer, 2012 [\[1\]](#)).

Even if all Minkowski tensors carry important geometric information, one type is particularly interesting and will be considered in the following:

$$\mathbf{W}_1^{0,p}(C) = \int_{\partial C} \underbrace{\mathbf{n} \odot \mathbf{n} \odot \dots \odot \mathbf{n}}_{p\text{-times}} d\partial C. \quad (3)$$

In this case, the symmetrized tensor product agrees with the classical tensor product. If applied to polygonal shapes the Minkowski tensors $\mathbf{W}_1^{0,p}(C)$ are related to the Minkowski problem for convex polytopes (i.e., generalizations of three-dimensional polyhedra to an arbitrary number of dimensions), which states that convex polytopes are uniquely described by the outer normals of the edges and the length of the corresponding edge (Minkowski, 1897 [\[2\]](#)). Several generalizations of this result exist (Klain, 2004 [\[3\]](#); Schneider, 2013 [\[4\]](#)), which makes the normal vectors a preferable quantity to describe shapes. However, Minkowski tensors contain redundant information, which asks for an irreducible representation. This can be achieved by decomposing the tensor with respect to the rotation group $SO(2)$ (Kapfer, 2012 [\[1\]](#); Mickel et al., 2013 [\[5\]](#); Klatt et al., 2022 [\[6\]](#)). Following this approach, one can write

$$\mathbf{W}_1^{0,p}(C) = \int_{S^1} \mathbf{u}^p \Psi_C(\mathbf{u}) d\mathbf{u} \quad \text{with} \quad \Psi_C(\mathbf{u}) = \int_{\partial C} \delta(\mathbf{n}(\mathbf{x}) - \mathbf{u}) d\partial C, \quad (4)$$

with $\Psi_C(\mathbf{u})$ being proportional to the probability density of the normal vectors. Identifying $\mathbf{u} \in S^1$ by the angle θ between $\mathbf{u}$ and the x -axis allows to write

$$\Psi_C(\theta) = \sum_{p=-\infty}^{\infty} \psi_p(C) e^{-ip\theta} \quad \text{with} \quad \psi_p(C) = \frac{1}{2\pi} \int_0^{2\pi} \Psi_C(\theta) e^{ip\theta} d\theta. \quad (5)$$

The Fourier coefficients $\psi_p(C)$ are the irreducible representations of $\mathbf{W}_1^{0,p}(C)$ and can be written as

$$\psi_p(C) = \frac{1}{2\pi} \int_{\partial C} e^{ip\theta_n} d\partial C, \quad (6)$$

where θ_n is the orientation of the outward pointing normal $\mathbf{n}$. The phase of the complex number $\psi_p(C)$ contains information about the preferred orientation and the absolute value $|\psi_p(C)|$ is a scalar index. We thus define

$$\vartheta_p(C) = \frac{1}{p} \arctan 2(\Im \psi_p(C), \Re \psi_p(C)) + \frac{\pi}{p} \quad \text{and} \quad q_p(C) = \frac{|\psi_p(C)|}{\psi_0(C)}, \quad (7)$$

with $\Im \psi_p$ and $\Re \psi_p$ the imaginary and real part of ψ_p , respectively, and $\psi_0(C) = \frac{W_1(C)}{2\pi}$. We have $q_p(C) \in [0, 1]$ quantifying the strength of p -atic order. **Figure 4** [\[7\]](#) illustrates the concept for polygonal and smooth shapes and **Figure 5** [\[8\]](#) illustrates the calculation of q_p and ϑ_p at the example of an equilateral triangle. Note that the constant factor of $1/(2\pi)$ in **Equation 6** [\[6\]](#) does not influence the result of **Equation 7** [\[7\]](#) and is therefore disregarded in **Figure 5** [\[8\]](#). Alternative derivations of **Equation 7** [\[7\]](#) consider higher order trace-less tensors (Virga, 2015 [\[9\]](#); Giomi et al., 2022 [\[10\]](#); Armengol-Collado et al., 2023 [\[11\]](#)) generated by the normal $\mathbf{n}$. In this case $\vartheta_p(C)$ (orange triatic director in **Figure 5** [\[8\]](#) right) correspond to the eigenvector to the negative eigenvalue and $\vartheta(C) - \pi/p$ (gray triatic director in **Figure 5** [\[8\]](#) right) to the eigenvector to the positive eigenvalue. For $p = 2$ this tensor based approach corresponds to the structure tensor (Mueller et al., 2019 [\[12\]](#)).

As pointed out in the introduction, p -atic orders have already been identified in biological tissue and model systems. However, the way to quantify rotational symmetry in these works differs from the Minkowski tensors. In Armengol-Collado et al. (2023) [\[11\]](#) the cell contour is approximated by a

Figure 4.

Regular and irregular shapes, adapted from Armengol-Collado et al. (2023) and Schaller et al. (2024), with magnitude and orientation calculated by Equation 7. For regular shapes, the corresponding magnitude of q_p is always 1.0 and the detected angle is the minimal angle of the p -atic orientation with respect to the x -axis. Note that no shape with $q_2 = 1.0$ is shown, as this would be a line. The visualization uses rotationally-symmetric direction fields (known as p -RoSy fields in computer graphics (Vaxman et al., 2016)). The brightness scales with the magnitude of the p -atic order, q_p , see color bars in Figure 1.

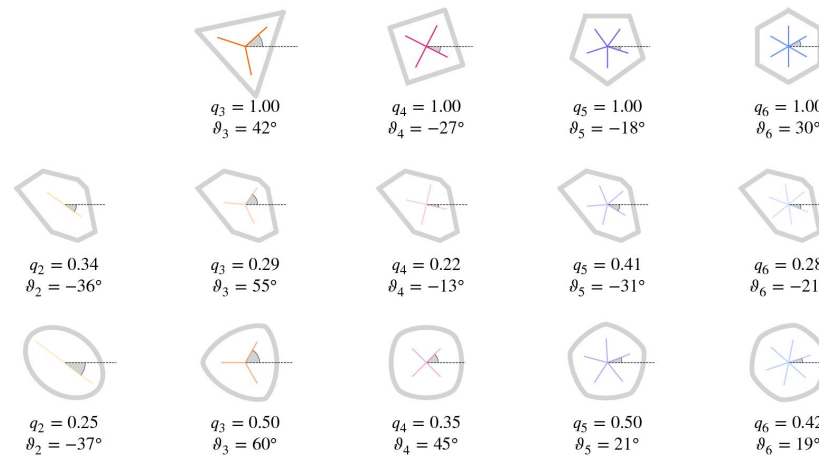
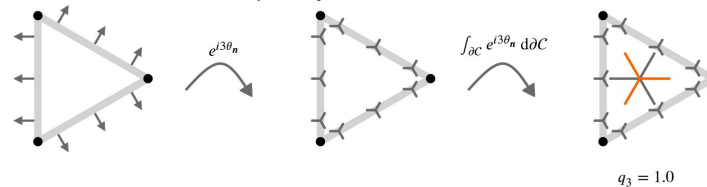


Figure 5.

Illustrative description of the definition of q_3 for an equilateral triangle. Considering rotational symmetries under a rotation $120^\circ = 2\pi/3$ means that vectors with an angle of 120° or 240° are treated as equal. Applied to the normals $\mathbf{n}$ (left), this means that under this rotational symmetry the normals on the three different edges are equal. Mathematically this is expressed through $e^{3i\theta_n}$ resulting in the triatic director shown instead of the normals $\mathbf{n}$ (middle). One leg of the triatic director always points in the direction as the normal. While only shown for 3 points on each edge, we obtain an orientation with the respective symmetry on every point of the contour $\partial\mathcal{C}$. Considering the line integral along the contour provides the dominating triatic director, shown in the center of mass (right). To get a value between 0.0 and 1.0 for q_3 we normalize this integral with the length of the contour, which corresponds to q_0 . As all triatic directors point in the same direction we obtain $q_3 = 1.0$ in this specific example. To be consistent with other approaches we rotate the resulting triatic director by $60^\circ = \pi/3$ leading to the orange triatic director, which is the quantity used for visualization.



polygon with V vertices having coordinates $\mathbf{x}_i$. The considered polygonal shape analysis is based on the shape function

$$\gamma_p(C) = \frac{\sum_{i=1}^V |\mathbf{r}_i|^p e^{ip\theta_i}}{\sum_{i=1}^V |\mathbf{r}_i|^p}, \quad (8)$$

with $\mathbf{r}_i = \mathbf{x}_i - \mathbf{x}_c$ being the vector from the center of mass $\mathbf{x}_c$ and θ_i being the orientation of the i -th vertex of the polygon with respect to its center of mass. As the center of mass can be computed by $\mathbf{x}_c = \frac{1}{V} \sum_{i=1}^V \mathbf{x}_i$ the only input data are the coordinates of the vertices of the V -sides polygon. Equation 8 captures the degree of regularity of the polygon by its amplitude $0 \leq |\gamma_p| \leq 1$ and its orientation ϑ_p which follows as

$$\vartheta_p(C) = \frac{1}{p} \arctan \frac{\Im \gamma_p(C)}{\Re \gamma_p(C)}, \quad (9)$$

with $\Im \gamma_p$ and $\Re \gamma_p$ the imaginary and real part of γ_p , respectively. Instead of the normals $\mathbf{n}$, the descriptor is based on the position vector $\mathbf{r}$. One might be tempted to think that Equation 9 is related to the irreducible representation of $\mathbf{W}_1^{p,0}$. This is, however, not the case as the weighting in Equation 9 is done with respect to the magnitude of $\mathbf{r}$, while in $\mathbf{W}_1^{p,0}$ the weighting is done with respect to the contour integral. Furthermore, in contrast to the Minkowski tensors, Equation 9 takes into account only the vectors at the vertices and not the vectors along the full cell contour. While this seems to be a technical detail, it has severe consequences. The theoretical basis, which guarantees continuity, no longer holds. This leads to unstable behaviour, as illustrated in Figure 6. While the shape of the polygons is almost identical, there is a jump in γ_3 and γ_4 as we go from four to three vertices. The weighting according to the magnitude cannot cure this, as the magnitude of the shrinking vector is far from zero shortly before this vertex vanishes. In the context of cellular systems, such deformations are a regular occurrence rather than an artificially constructed test case. We will demonstrate the impact while discussing the results and recommend only using robust descriptors such as the Minkowski tensors or their irreducible representations.

We further note that bond order parameters Ψ_p^{bond} , which consider the connection between cells, have also been used as shape descriptors (Li and Ciamarra, 2018; Durand and Heu, 2019; Pasupalak et al., 2020). Introduced in Nelson and Halperin (1979), these parameters are

$$\Psi_p^{bond} = \frac{1}{B} \sum_{b=1}^B e^{ip\theta_b}, \quad (10)$$

with B denoting the number of bonds and θ_b the orientation of the bond b . In the context of mono-layer tissues, B is understood as the number of neighbors, and θ_b as the orientation of the connection of the center of mass of the current cell with the center of mass of the neighbor b (Loewe et al., 2020; Monfared et al., 2023). So cells with the same contour but different neighbor relations lead to different bond order parameters. This characteristic should already disqualify these measures as shape descriptors. However, as discussed in detail in Mickel et al. (2013), they are not robust even for the task they are designed for. We therefore do not discuss them further.

Simulation methods

We consider two modeling approaches, an active vertex model and a multiphase field model. Both have been proven to capture various generic properties of epithelial tissue (Hakim and Silberzan, 2017; Alert and Treppe, 2020; Moure and Gomez, 2021). The key aspects of formulations are out-lined in Box 1 and Box 2. They refer to (Koride et al., 2018; Das et al., 2021; Killeen et al., 2022) and (Wenzel and Voigt, 2021; Jain et al., 2023, 2024b), respectively. Parameters used within these models are listed in Appendix 1 Table 1 and Appendix 1 Table 2.

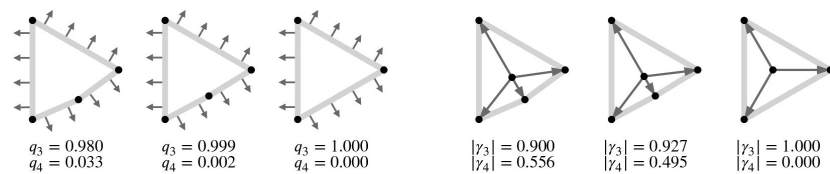


Figure 6.

Defining p -atic order for deformable objects requires robust shape descriptors. Shown is the strength of p -atic order for a polygon converging to an equilateral triangle. a) using q_p and b) using γ_p . The considered vectors used in the computations, normals $\mathbf{n}$ of the contour for the Minkowski tensors and $\mathbf{r}_i$ for γ_p , are shown. Note that the choice of the center of mass highly influences the value of γ_p . We here used the described approach following [Armengol-Collado et al. \(2023\)](https://doi.org/10.7554/eLife.105680.1).

The initial configuration for the active vertex model simulations was built by placing N points randomly in a periodic simulation box of length $L \times L$ and ensuring that no two points were within a distance less than r_{cut} from each other. The points were used to create a Voronoi tiling. Typically, such a tiling had cells of very irregular shapes. To make cell shapes more uniform, the centroids of each tile were computed and used as seeds for a new Voronoi tiling. This process was iterated until it converged within the tolerance of 10^{-5} , resulting in a so-called well-centered Voronoi tiling with cells of random shapes but similar sizes. Initial directions of polarity vectors $\hat{\mathbf{n}}_C$ were chosen at random from a uniform distribution. The initial configuration for the multiphase field model simulations considers a regular arrangement of N cells of equal size in a periodic simulation box of length $L \times L$ with randomly chosen directions of self-propulsion and simulating for several time steps until the cell shapes appear sufficiently irregular. For both models we consider 100 cells and analyse time instances of the evolution.

Experimental setup

Cell culture

Madin-Darby canine kidney (MDCK) cells were cultured in DMEM (DMEM, low glucose, GlutaMAX™ Supplement, pyruvate) supplemented with 10 % fetal bovine serum (FBS; Gibco) and 100 U/mL peni-cillin/streptomycin (Gibco) at 37 °C with 5 % CO₂. The cell line was tested mycoplasma.

Monolayer preparation

Cells were seeded on glass-bottom dishes (Mattek) pretreated with 10 µg/mL fibronectin human plasma in phosphate buffered saline (PBS, pH 7.4; Gibco). Fibronectin was incubated for 30 min at 37 °C. The initial cell seeding density was sparse. They were imaged approximately 24 hours after, when a confluent monolayer was formed.

Box 1. Active vertex model

The active vertex model is based on models discussed in (Koride et al., 2018 [↗](#); Das et al., 2021 [↗](#); Killeen et al., 2022 [↗](#)). A confluent epithelial tissue is appreciated as a two-dimensional tiling of a plane with the elastic energy given as (Honda, 1983 [↗](#); Farhadifar et al., 2007 [↗](#); Honda and Nagai, 2022 [↗](#))

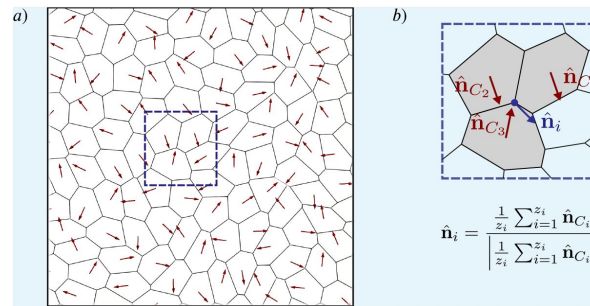
$$E_{\text{VM}} = \frac{K_A}{2} \sum_C (A_C - A_0)^2 + \frac{K_P}{2} \sum_C (P_C - P_0)^2, \quad (11)$$

where A_C and P_C are, respectively, area and perimeter of cell C , A_0 and P_0 are, respectively, preferred area and perimeter, and K_A and K_P are, respectively, area and perimeter moduli. For simplicity, it is assumed that all cells have the same values of A_0 , P_0 , K_A , and K_P . The model can be made dimensionless by dividing Equation 11 [↗](#) by $e^* = K_A A_0^2$,

$$e_{\text{VM}} = \frac{1}{2} \sum_C (a_C - 1)^2 + \frac{k_P}{2} \sum_C (p_C - p_0)^2, \quad (12)$$

where $a_C = A_C/A_0$ (i.e. $l^* = \sqrt{A_0}$ is the unit of length), $p_C = P_C/\sqrt{A_0}$, $k_P = K_P/(K_A A_0)$, and $p_0 = P_0/\sqrt{A_0}$. p_0 is called the shape index, and it has been shown to play a central role in determining if the tissue is in a solid or fluid state (Bi et al., 2015 [↗](#)).

One can use Equation 11 [↗](#) to find the mechanical force on a vertex i as $\mathbf{F}_i = -\nabla_{\mathbf{r}_i} E_{\text{VM}}$, where $\nabla_{\mathbf{r}_i}$ is the gradient with respect to the position $\mathbf{r}_i$ of the vertex. The expression for $\mathbf{F}_i$ is given in terms of the position of the vertex i and its immediate neighbours (Tong et al., 2023 [↗](#)), which makes it fast to compute. Furthermore, cells move on a substrate by being noisily self-propelled along the



Box 1—figure 1.

a) Cell contour of the active vertex model. Red arrows represent the polarity vectors that set each cell's instantaneous direction of self-propulsion. b) Zoom in on a vertex surrounded by three cells showing how the direction of self-propulsion on a vertex is calculated.

direction of their planar polarity described by a unit-length vector $\hat{\mathbf{n}}_i$ from which the polarity direction at the vertex $\hat{\mathbf{n}}_i$ is defined, see **Box 1 - figure 1 b**, where z_i is the number of neighbours of vertex i . It is convenient to write $\hat{\mathbf{n}}_i = \cos(\alpha_i)\mathbf{e}_x + \sin(\alpha_i)\mathbf{e}_y$, where α_i is an angle with the x -axis of the simulation box. The equations of motion for vertex i is a force balance between active, elastic, and frictional forces and are given as

$$\dot{\mathbf{r}}_i = v_0 \hat{\mathbf{n}}_i + \frac{1}{\gamma_{fr}} \mathbf{F}_i, \quad \dot{\alpha}_i = \xi_i(t), \quad (13)$$

where v_0 is the magnitude of the self-propulsion velocity defined as f_0/γ_{fr} . Furthermore the overdot denotes the time derivative, γ_{fr} is the friction coefficient, f_0 is the magnitude of the self-propulsion force (i.e., the activity), $\xi_i(t)$ is Gaussian noise with $\langle \xi_i(t) \rangle = 0$ and $\langle \xi_i(t) \xi_j(t') \rangle = 2D_r \delta_{ij} \delta(t - t')$, where D_r is the rotation diffusion constant and $\langle \cdot \rangle$ is the ensemble average over the noise. Equations of motion can be non-dimensionalised by measuring time in units of $t^* = \gamma_{fr}/(K A)$ and force in units of $f^* = K_A A_0^{3/2}$ and integrated numerically using the first-order Euler-Maruyama method with timestep τ .

Box 2. Multiphase field model

The multiphase field modelling approach follows Wenzel and Voigt (2021); Jain et al. (2023, 2024b). Each cell is described by a scalar phase field variable ϕ_i with $i = 1, 2, \dots, N$, where the bulk values $\phi_i \approx 1$ denotes the cell interior, $\phi_i \approx -1$ denotes the cell exterior, and with a diffuse interface of width (ϵ) between them representing the cell boundary. The phase field ϕ_i follows the conservative dynamics

$$\partial_t \phi_i + \mathbf{v}_i \cdot \nabla \phi_i = \Delta \frac{\delta \mathcal{F}}{\delta \phi_i}, \quad (14)$$

with the free energy functional $\mathcal{F} = \mathcal{F}_{CH} + \mathcal{F}_{INT}$ containing the Cahn-Hilliard energy

$$\mathcal{F}_{CH} = \frac{1}{Ca} \sum_{i=1}^N \int_{\Omega} \left(\frac{\epsilon}{2} \|\nabla \phi_i\|^2 + \frac{1}{4\epsilon} (\phi_i^2 - 1)^2 \right) d\mathbf{x}, \quad (15)$$

where the Capillary number Ca is a parameter for tuning the cell deformability, and an interaction energy consisting of repulsive and attractive parts defined as

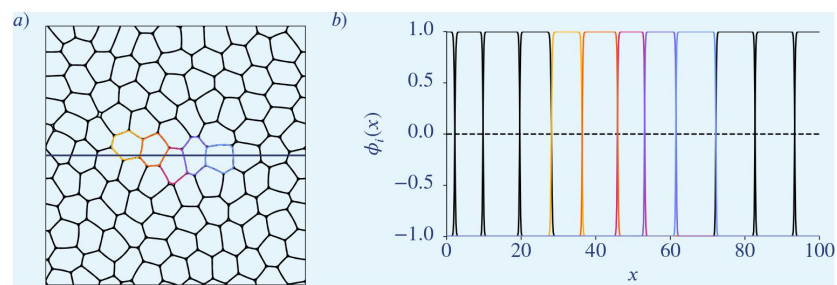
$$\mathcal{F}_{INT} = \frac{1}{In} \sum_{i=1}^N \int_{\Omega} \sum_{j \neq i} \left(a_r (\phi_i + 1)^2 (\phi_j + 1)^2 - a_a (\phi_i^2 - 1)^2 (\phi_j^2 - 1)^2 \right) d\mathbf{x}, \quad (16)$$

where In denotes the interaction strength, and a_r and a_a are parameters to tune contribution of repulsion and attraction. The repulsive term penalises overlap of cell interiors, while the attractive part promotes overlap of cell interfaces. The relation to former formulations is discussed in Happel and Voigt (2024).

Each cell is self-propelled, and the cell activity is introduced through an advection term. The cell velocity field is defined as

$$\mathbf{v}_i(\mathbf{x}, t) = v_0 \mathbf{e}_i(t) \hat{\phi}_i(\mathbf{x}, t), \quad (17)$$

where v_0 is used to tune the magnitude of activity, $\hat{\phi}_i = \frac{\phi_i + 1}{2}$ $\mathbf{e}_i = [\cos \theta_i(t), \sin \theta_i(t)]$ is its direction. The migration orientation $\theta_i(t)$ evolves diffusively with a drift that aligns to the principal axis of cell's elongation as $d\theta_i = \sqrt{2D_r} dW_i(t) + \alpha(\beta_i(t) - \theta_i(t))dt$, where D_r is the rotational diffusivity, W_i is the Wiener process, $\beta_i(t)$ is the orientation of the cell elongation and α controls the time scale of this alignment.



Box 2—figure 1.

a) Cell contours of the multiphase field model. b) Corresponding phase field functions along the horizontal line in a). Colours correspond to the once in a).

The resulting system of partial differential equations is considered on a square domain with periodic boundary conditions and is solved by the finite element method within the toolbox AMDiS (Vey and Voigt, 2007 [DOI](#); Witkowski et al., 2015 [DOI](#)) and the parallelization concept introduced in Praetorius and Voigt (2018) [DOI](#) is considered, which allows scaling with the number of cells. We in addition introduce a de Gennes factor in $\mathcal{F}_{CH}$ to ensure $\Phi_i \in [-1, 1]$, see Salvalaglio et al. (2021) [DOI](#).

Live cell imaging

Samples were imaged using Nikon ECLIPSE Ti microscope equipped with a H201-K-FRAME Okolab chamber, heating system (Okolab) and a CO₂ pump (Okolab) which maintained them at 37 °C and at 5 % CO₂. Phase microscopy images were taken each 10th minute using a 10× NA=0.3 Plan Fluor objective and Andor Neo 5.5 sCMOS camera.

Image analysis

The time-series were xy-drift corrected using Fast4DReg plugin Laine et al. (2019) [DOI](#), Pylvänäinen et al. (2022) [DOI](#) in FIJI. Cells were segmented and tracked using the python module CellSegmentation-Tracker (<https://github.com/simonguld/CellSegmentationTracker> [DOI](#)), which utilizes both Cellpose Stringer et al. (2021) [DOI](#) and Trackmate Tinevez et al. (2017) [DOI](#). A Cellpose model was trained by manual segmentation of the phase contrast images.

Results

Quantifying orientational order in biological tissues can be realized by Minkowsky tensors. The orientation ϑ_p and the strength of p -atic order q_p in Equation 7 [DOI](#) can be computed for any integer p . These quantities describe how cell shapes align with specific rotational symmetries. A recurring challenge in the field is determining whether q_p values for different p can be meaningfully compared. This question is particularly important for interpreting nematic (q_2) and hexatic (q_6) orders, which describe distinct symmetries but often coexist in biological systems. However, a direct comparison is problematic as q_2 and q_6 reveal significant differences in their ranges: $q_2 = 1.0$ represents an idealized line shape that is unattainable for cells, while $q_6 = 1.0$ corresponds to a perfect hexagon observed in experiments and simulations. To further address this, we analyze simulation data from two computational models, examining how deformability and activity—key determinants of tissue mechanics—affect q_p values and their interplay. We employ the active vertex model and the multiphase field model, both of which incorporate activity and deformability. These two parameters are recognized as crucial for capturing the coarse-grained properties of confluent tissues (Jain et al., 2023, 2024b [DOI](#)). By leveraging these complementary frameworks, we ensure that our conclusions are robust across model-specific assumptions and parameter regimes. This dual approach allows us to systematically investigate how these models capture tissue behavior, particularly regarding cell shape variability and resulting p -atic orders.

In the active vertex model, deformability is controlled by the target shape index p_0 , reflecting the balance between cell-cell adhesion and cortical tension. The multiphase field model, by contrast, encodes deformability through the capillary number Ca , which directly incorporates cortical tension. Although these models differ conceptually, both have been validated in studies of cell monolayer mechanics, including solid-liquid transitions, neighbor exchange dynamics, and stress profiles (Fletcher et al., 2014 [DOI](#); Alt et al., 2017 [DOI](#); Li et al., 2019 [DOI](#); Balasubramaniam et al., 2021 [DOI](#); Wenzel and Voigt, 2021 [DOI](#); Sknepnek et al., 2023 [DOI](#); Melo et al., 2023 [DOI](#)). Importantly, the active vertex model enforces confluency by design, while the multiphase field model approximates it through dynamic cell-cell interactions, leading to differences in topological rearrangements and cell shape representations.

We vary activity strength (v_0), the shape index (p_0), and the capillary number (Ca), ensuring all parameter combinations remain in the fluid regime. Fluidity was confirmed using metrics such as the mean square displacement (MSD) (Loewe et al., 2020 [DOI](#)), neighbor number variance (Wenzel and Voigt, 2021 [DOI](#)), and the self-intermediate scattering function (Bi et al., 2016 [DOI](#)). Using Minkowski tensors, we computed ϑ_p and q_p values as described in Equation 7 [DOI](#), focusing on $p = 2$ and $p = 6$ due to their biological relevance. In solid-like states, such as hexagonal cellular arrangements, hexatic order ($p = 6$) is the appropriate descriptor, as hexagons exhibit $60^\circ = 2\pi/6$ rotational symmetry. Hexatic order has been extensively studied in biological systems (Li and Ciamarra, 2018 [DOI](#); Durand and Heu, 2019 [DOI](#); Pasupalak et al., 2020 [DOI](#)), and its importance in cellular organization is well-documented. In fluid-like states, cells often become elongated, exhibiting nematic order ($p = 2$), characterized by $180^\circ = 2\pi/2$ rotational symmetry. Elongated cell shapes are also common in specific cell types, such as fibroblasts. Quantifying nematic order using q_2 is particularly relevant, as nematic configurations and defects have been linked to critical biological processes, including hydra morphogenesis (Maroudas-Sacks et al., 2021 [DOI](#)) and cell death (Saw et al., 2017 [DOI](#)).

Independence of q_2 and q_6

Recent studies suggest a hexatic-nematic crossover in epithelial tissues, where hexatic order dominates at small scales and nematic order prevails at larger scales (Eckert et al., 2023 [DOI](#); Armengol-Collado et al., 2023 [DOI](#), 2024 [DOI](#)). To explore this relationship, we examine the distribution of (q_2 , q_6) values across deformability-activity parameter pairs in both models (Figure 7 [DOI](#)). Despite structural differences between the models, both show consistent trends. Closer examination shows that in near-solid regimes (Figure 7 [DOI](#) lower left), q_2 and q_6 values cluster tightly due to restricted shape fluctuations. However, even in this regime, small q_2 values can correspond to either small or large q_6 values, and vice versa. In more fluid-like regimes, with higher activity and higher deformability (Figure 7 [DOI](#) upper right), q_2 and q_6 values become highly scattered. Each q_2 value spans a broad range of q_6 values, and vice versa, demonstrating their independence. These findings underscore that q_p values for different p measure distinct aspects of cell shape anisotropy and cannot be meaningfully compared directly.

This conclusion also holds on larger length scales. In order to test the hypothesis of a hexatic-nematic crossover at larger length scales (Armengol-Collado et al., 2023 [DOI](#)) we consider a coarse-graining of q_2 and q_6 across spatial domains. Details are described in Appendix 2 [DOI](#).

Appendix 2 - Figure 13 [DOI](#) and Appendix 2 - Figure 14 [DOI](#) show the results for the active vertex model and the multiphase field model, respectively. The results show no consistent trends indicative of a crossover, regardless of the considered model and coarse-graining radius. Instead, the apparent convergence of Q_2 and Q_6 values (coarse-grained values of q_2 and q_6) with increasing activity or deformability simply reflects the same trends observed at the individual cell scale.

Dependence of q_p on activity and deformability

The previous analysis demonstrated that comparing q_p across different values of p is not meaningful. However, fixing p and exploring q_p under varying physical conditions offers valuable insights into how tissue properties such as activity and deformability influence cell shape and orientational order. We again focus on nematic ($p = 2$) and hexatic ($p = 6$), as shown in Figure 8 [DOI](#) and Figure 9 [DOI](#). Additional results for $p = 3, 4, 5$ are provided in Figure 8—figure Supplement 1 [DOI](#)—Figure 8—figure Supplement 3 [DOI](#) (fixed activity) and Figure 9—figure Supplement 1 [DOI](#)—Figure 9—figure Supplement 3 [DOI](#) (fixed deformability). In all plots all cells and all time steps are considered. In Figure 8 [DOI](#), we vary deformability (p_0 in the active vertex model and Ca in the multiphase field model) while keeping the activity v_0 constant. Results are presented for the active vertex model (left column) and the multiphase field model (right column). Activity increases from bottom to top rows. Both models show qualitatively similar trends in the probability density functions (PDFs) of q_2 and q_6 . For $p = 6$, increasing deformability shifts the PDF of q_6 to the left,

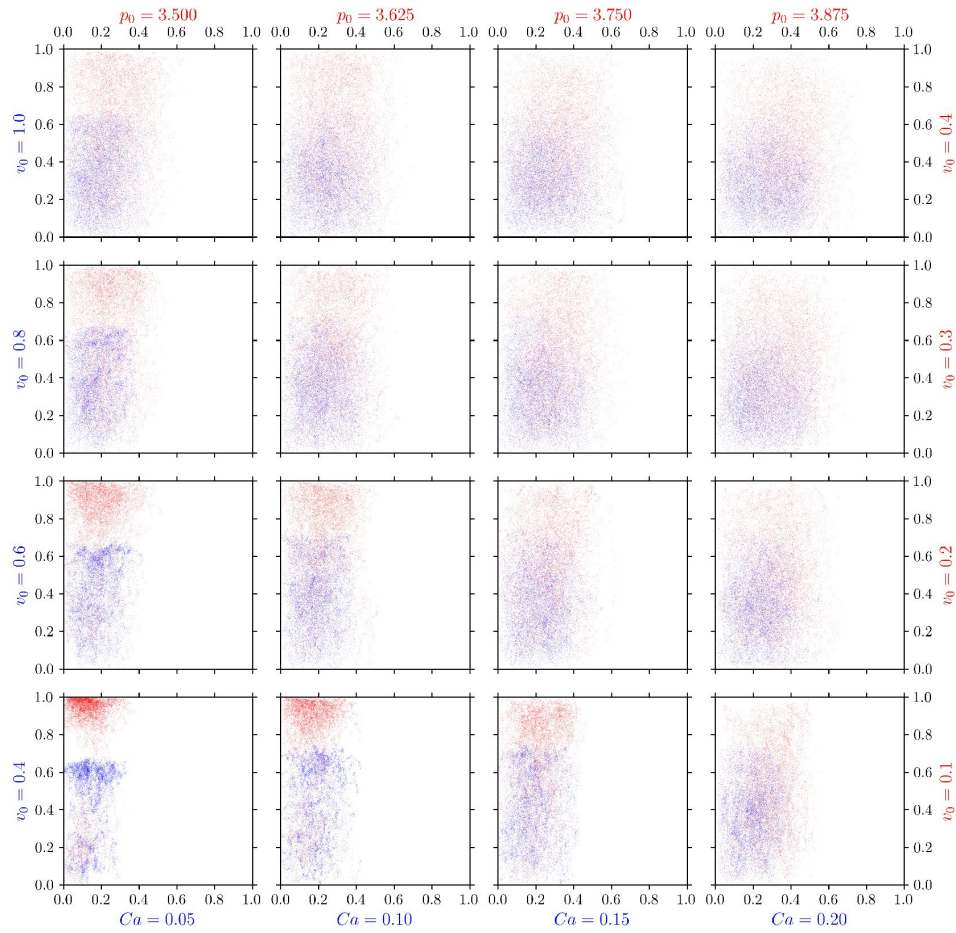


Figure 7.

Nematic ($p = 2$) and hexatic ($p = 6$) order are independent of each other. q_6 (y-axis) versus q_2 (x-axis) for all cells in the multiphase field model (blue) and active vertex model (red). For each cell and each timestep we plot one point (q_2, q_6). Each panel corresponds to specific model parameters; Ca and v_0 for multiphase field model, and p_0 and v_0 for the active vertex model, representing deformability and activity, respectively.

indicating lower mean values. In contrast, for $p = 2$, higher deformability leads to higher q_2 values, reflected in a rightward shift of the PDF. This trend is confirmed by the mean values, shown as a function of deformability (p_0 and Ca , respectively) in the inlets. Additionally, the PDFs broaden with increasing deformability, and this effect is more pronounced at lower activity levels. A notable difference between the models is the range of q_p values. In the multiphase field model, q_p rarely exceeds 0.8 due to the smoother, more rounded cell shapes, whereas the active vertex model often produces higher q_p values. In **Figure 9**, deformability is held constant while activity v_0 is varied. Results are again shown for the active vertex model (left column) and the multiphase field model (right column), with increasing activity indicated by brighter colors. Deformability increases from bottom to top rows. The trends mirror those observed in **Figure 8**. Increasing activity reduces the mean value of q_6 while increasing q_2 , see inlets. The effects of activity are more pronounced at lower deformabilities; at higher deformabilities, differences between parameter regimes diminish. The results agree qualitatively with the ones obtained for the measures $|\gamma_2|$ and $|\gamma_6|$ reported in (Armengol-Collado et al., 2023, Extended Data Fig. 3). Note that in the calculation of γ_p every cell is approximated by a polygon. Therefore, also for the multiphase field model the obtained values range up to 1.0. The overall behavior of q_2 and q_6 is summarized in **Figure 10**, which shows the mean values $\overline{q_2}$ and $\overline{q_6}$ as functions of activity and deformability. For both models:

- $\overline{q_2}$ increases with higher activity or deformability
- $\overline{q_6}$ decreases with higher activity or deformability.

This behaviour is consistent with a trend towards nematic order in more dynamic regimes and toward hexagonal order in more constrained, less dynamic regimes. The first is further confirmed by recent findings in analysing T1 transitions and their effect on cell shapes Jain et al. (2024a). These studies suggest that cells transiently elongate when they are undergoing T1 transitions. As the number of T1 transitions increases with activity or deformability Jain et al. (2024a) this elongation contributes to the observed behavior. The second is consistent with the emergence of hexagonal arrangements in solid-like states. Corresponding results for $p = 3, 4, 5$ are shown in **Figure 10—figure Supplement 1**.

Comparison with experiments

Next, we apply shape descriptor framework to experiments on epithelial cells. We consider a monolayer of Madine-Derby-Canine-Kidney (MDCK) cells. We consider two types of cell monolayers: MDCK wild type cells and MDCK E-cad KO cells, which have the E-cadherin protein removed. These E-cad KO cells have weaker cell-cell adhesion strengths and previous studies have shown that compared to wild-type cells, the E-cad KO cells have lower velocities, same spreading area, higher aspect ratio, and higher stiffness Balasubramaniam et al. (2021). The slower movement of E-cad KO cells compared to the wild-type cells suggest their lower activity, and their higher stiffness can be interpreted as having lower deformability compared to the wild-type cells. However, care must be taken in directly comparing the two cell types and directly comparing to simulation results, as the key parameters varied in simulations—cell activity and deformability—are difficult to manipulate in isolation in experiments. One complicating factor is the strong dependence of cell shape on cell density, as previously reported (Eckert et al., 2023). Higher cell densities generally lead to rounder cell shapes, which we confirmed in two distinct experimental setups (see **Figure 11**). For both wild-type MDCK cells and E-cadherin knockout variants, increased cell density corresponded to lower mean values of q_2 , consistent with rounder cell geometries. For higher values of q_p , however, this density-dependent effect was less pronounced. A plausible explanation lies in the geometry of shapes for larger q_p . Rounded shapes close to a circle can yield relatively high q_p values for certain p , such as the example shown in the third row of **Figure 4**. Distinguishing such shapes from perfect circles requires highly accurate data, which can be difficult to obtain in experimental imaging, potentially masking density-dependent effects for $p = 6$.

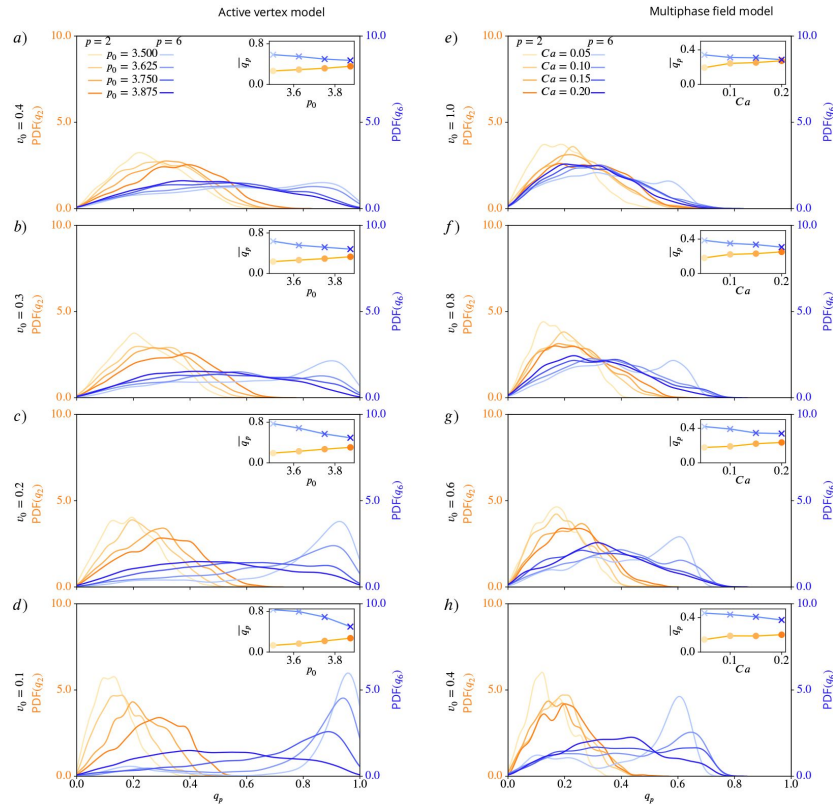


Figure 8.

Nematic ($p = 2$) and hexatic ($p = 6$) order depend on deformability of the cells. Probability distribution functions (PDFs) for q_2 (shades of orange) and q_6 (shades of blue), using kde-plots, for varying deformability p_0 or Ca and fixed activity v_0 . Insets show mean values of q_2 and q_6 as function of deformability. a) - d) active vertex model, e) - h) multiphase field model for decreasing activity.

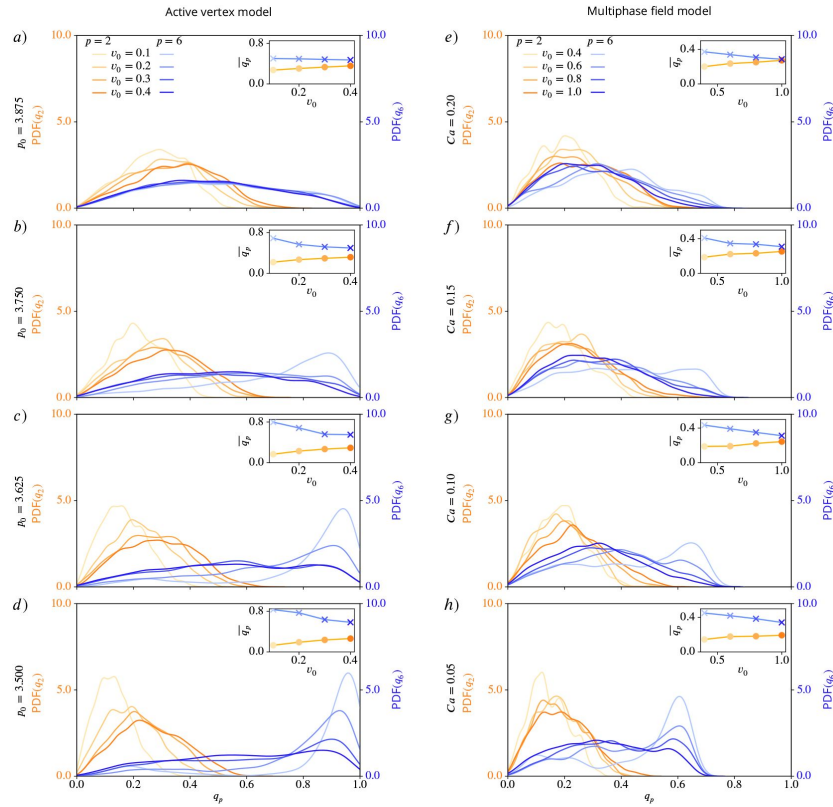


Figure 9.

Nematic ($p = 2$) and hexatic ($p = 6$) order depend on activity of the cells. Probability distribution functions (PDFs) for q_2 (shades of orange) and q_6 (shades of blue), using kde-plots, for varying activity v_0 and fixed deformability p_0 or Ca . Insets show mean values of q_2 and q_6 as function of activity. a) - d) active vertex model and e) - h) multiphase field model for decreasing deformability.

Figure 10.

Nematic ($p = 2$) and hexatic ($p = 6$) order depend on activity and deformability of the cells. Mean value $\overline{q_p}$ for $p = 2$ (left) and $p = 6$ (right) as function of deformability p_0 or Ca and activity v_0 for active vertex model (a) and b)) and multiphase field model (c) and d)).

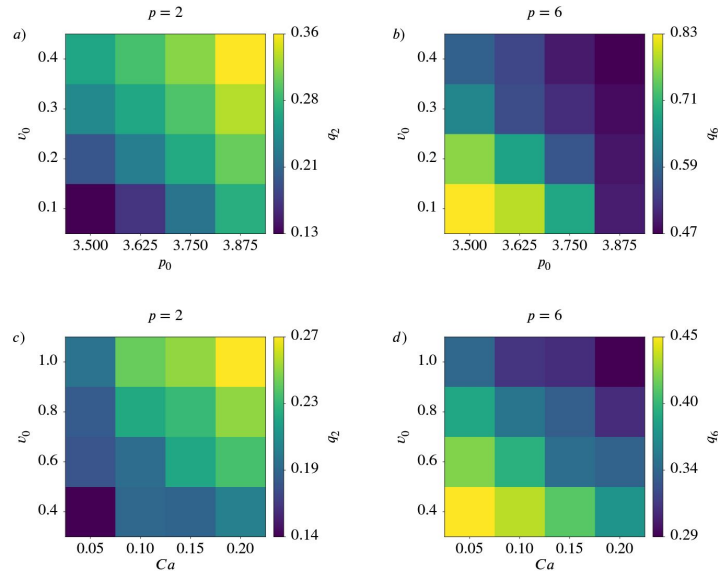
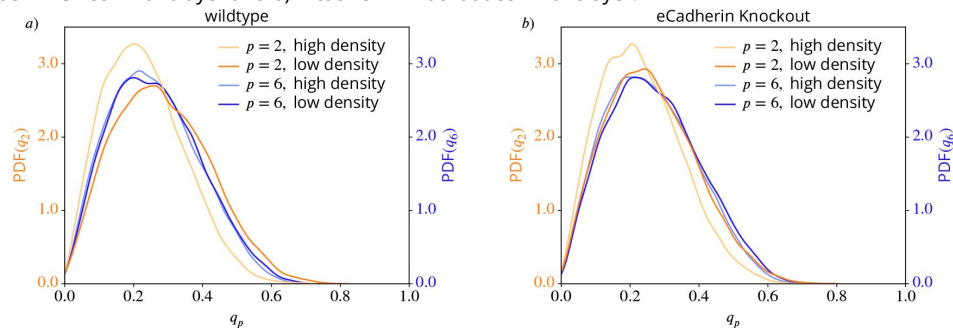


Figure 11.

In experimental system of MDCK cells, nematic ($p = 2$) and hexatic ($p = 6$) order respond differently to change in cell density. Probability distribution functions (PDFs) using kde-plots, for q_2 (orange shaded) and q_6 (blue shaded) for different densities in a) wildtype MDCK cell monolayer and b) E-cadherin knockout cell monolayer.



Importantly, due to the substantial differences in cell density between wild-type cells and E-cadherin knockout variants, direct comparisons between these two experimental conditions would be unjustifiable. Density influences cell shape too strongly to isolate other effects in these datasets. Despite these limitations, we observed qualitative agreement between experimental and computational results regarding the independence of q_2 and q_6 (see **Figure 12**). Both measures appear to be unrelated, reinforcing the conclusion that nematic ($p = 2$) and hexatic ($p = 6$) order describe distinct aspects of cell shape and alignment. Consequently, comparing q_2 and q_6 directly is not meaningful, whether in simulations nor in experiments.

Sensitivity of the results on the considered shape descriptor

In order to demonstrate that other methods, such as polygonal shape analysis [Armengol-Collado et al. \(2023\)](#) fail to capture the nuances of irregular cell shapes we also use [Equation 8](#) as a shape descriptor for the considered datasets. For the experimental data and the multiphase field model this first requires an approximation of the cell shape by a polygon. For finding the vertices of the polygon we consider the Voronoi interface method ([Saye and Sethian, 2011](#), [2012](#)), see [Appendix 3](#). The corresponding figures to **Figure 1**, **Figure 2** and **Figure 4** are shown in **Appendix 3 Figure 16**, **Appendix 3 Figure 17** and **Appendix 3 Figure 15**, respectively. While already these results show quantitative differences, the failure of the approach can best be seen in the qualitative difference between **Figure 10** and the corresponding results in **Appendix 3 Figure 18**, which, instead of the monotonic trend for activity and deformability, exhibit nonmonotonic trends, peaking at intermediate values of activity or deformability. These divergences highlight the limitations of γ_p in capturing consistent patterns and emphasize the need for stable, reliable shape descriptors like the Minkowski tensors.

For completeness we also provide the corresponding figures to **Figure 7** and **Figure 13** in **Appendix 3 Figure 19** and **Appendix 3 Figure 20**, respectively. We hereby only consider the active vertex model. The results are similar to the once obtained using the Minkowski tensors and confirm the independence of p -atic order for different symmetries.

Discussion

In this study, we introduced Minkowski tensors as a robust and versatile tool for quantifying p -atic order in multicellular systems, particularly in scenarios involving rounded or irregular cell shapes. By applying this framework to extensive datasets from two distinct computational models—the active vertex model and the multiphase field model—we identified universal trends: increasing activity and deformability of the cells enhance nematic order ($p = 2$) while diminishing hexatic order ($p = 6$). The consistency of these findings across two models, despite their inherent differences, underscores the generality and relevance of our results.

While various shape characterization methods, such as the bond order parameter ([Loewe et al., 2020](#); [Monfared et al., 2023](#)) and the shape function γ_p ([Armengol-Collado et al., 2023](#)), have been explored in the literature, we demonstrated that the choice of shape descriptor significantly impacts the conclusions drawn. Applying γ_p to data from the active vertex model revealed qualitative discrepancies: for $p = 2$, γ_p and q_p behaved similarly, but for $p = 6$, γ_p exhibited significant changes. Such divergences, together with limited mathematical foundations, highlight the limitations of γ_p in capturing consistent patterns and emphasize the need for stable, reliable shape measures like the Minkowski tensors. In contrast to the bond order parameter or the shape function γ_p the stability of Minkowski tensors can be mathematically justified. We have further demonstrated why bond order parameter or the shape function γ_p fail in certain situations. This finding is not merely a technical nuance; the choice of shape detection method fundamentally

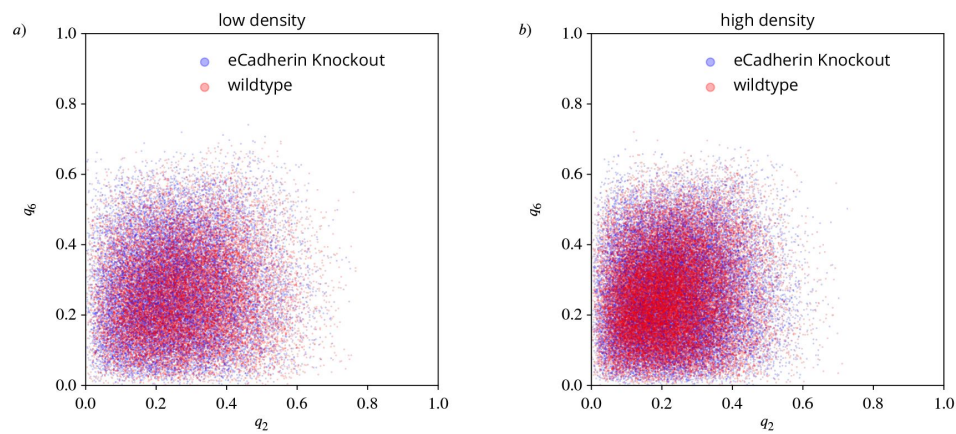


Figure 12.

In experimental system of MDCK cells nematic ($p = 2$) and hexatic ($p = 6$) order are independent of each other for different cell densities and different cell-cell adhesion strengths. q_6 (y-axis) versus q_2 (x-axis) for different densities a) low density and b) high density, for wildtype MDCK cell monolayer and E-cadherin knockout cell monolayer. For each cell and each timestep we plot one point (q_2, q_6) . The E-cadherin cells lack an adhesion protein and have therefore weaker cell-cell adhesions.

influences the interpretation of p -atic order. Consequently, selecting a robust shape descriptor is critical for obtaining meaningful and reproducible insights into orientational order in multicellular systems.

A critical question in the literature has been whether shape measures for different p -atic orders can be directly compared. While some studies have suggested relationships between γ_2 and γ_6 (Armengol-Collado et al., 2023 [DOI](#)), our results refute this notion. We demonstrated that measures like q_2 and q_6 capture fundamentally distinct aspects of cell shape and alignment, and comparing them directly is both mathematically and biologically misleading. We further tested the hypothesis of a hexatic-nematic crossover at larger length scales by coarse-graining q_2 and q_6 across spatial domains. The results showed no consistent trends indicative of a crossover, regardless of the coarse-graining strategy and the considered model. Instead, the apparent convergence of Q_2 and Q_6 values (coarse-grained values of q_2 and q_6) with increasing activity or deformability simply reflected the same trends observed at the individual cell scale. This reinforces the conclusion that comparisons of p -atic orders across different p lack a robust basis and that claims of crossovers require careful reconsideration.

Our findings suggest that p -atic orders should be studied independently across length scales, as they describe complementary aspects of cellular organization. The coexistence of distinct orientational orders emphasized in different studies—such as nematic ($p = 2$) (Duclos et al., 2017 [DOI](#); Saw et al., 2017 [DOI](#); Kawaguchi et al., 2017 [DOI](#)), tetratic ($p = 4$) (Cislo et al., 2023 [DOI](#)), and hexatic ($p = 6$) (Li and Ciamarra, 2018 [DOI](#))—is not contradictory but highlights the rich, multifaceted nature of cellular organization. Rather than searching for a single dominant order, future research should focus on the interplay of different p -atic orders and their associated defects. This suggests to not only consider p -atic liquid crystal theories Giomi et al. (2022) [DOI](#) for one p , but combinations of these models for various p 's. Understanding how these orders interact may reveal how they collectively regulate morphogenetic processes.

Connecting p -atic orders to biological function remains a critical avenue for exploration. While the mathematical independence of q_2 , q_6 , and other shape measures precludes the identification of a universal dominant order, biological systems may exhibit context-dependent preferences. For example, a specific p -atic order might correlate with or drive a key morphogenetic event. Investigating these connections could yield insights into how tissues achieve functional organization and adapt to environmental cues. As such, while it is difficult to speak of dominating orders from a mathematical point of view, there could be a dominating order from a biological point of view, meaning the p -atic order connected to the governing biological process.

Note

This reviewed preprint has been updated to correct the image for Figure 1, which was previously a duplicate of Figure 8 due to an error at the preprint server.

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“Vector- and tensor-valued surface PDEs”) and computing resources provided by JSC through MORPH and by ZIH through WIR. Views and opinions expressed are however those of the authors only and do not necessarily reflect those of the European Union or the European Research Council. Neither the European Union nor the granting authority can be held responsible for them.

Appendix 1

Parameters for the computational models

Appendix 2

Coarse graining results obtained with the Minkowski tensor

For comparison with the literature we are considering a coarse-graining, following closely the strategy used in Armengol-Collado et al. (2023) [\[1\]](#). We therefore regard the coarse-grained strength of p -atic order $Q_p = Q_p(\mathbf{x})$, which is the average of all shape functions $\frac{q_p}{q_0}$ (or equivalently $q_p e^{ip\theta_p}$) of cells whose center of mass $\mathbf{x}_i$ lies within a circle with radius R and center $\mathbf{x}$. In a formula this is:

$$Q_p(\mathbf{x}) = \frac{\sum_{j=1}^{100} q_p(C_j) e^{ip\theta_p(C_j)} \Theta(R - |\mathbf{x} - \mathbf{x}_c(C_j)|)}{\sum_{j=1}^{100} \Theta(R - |\mathbf{x} - \mathbf{x}_c(C_j)|)} \quad (18)$$

whereby the formula uses that our simulations contain 100 cells. $\mathcal{C}_i$ denotes the i -th cell and Θ denotes the Heaviside step function with $\Theta(x) = 1$ for $x > 0$ and $\Theta(x) = 0$ otherwise. As in Armengol-Collado et al. (2023) [\[1\]](#) the position $\mathbf{x}$ is sampled over a square grid with spacing $R_c = \sqrt{\frac{A_c}{\pi}}$ with A_c the cell area. For **Appendix 2 Figure 13** [\[2\]](#) and **Appendix 2 Figure 14** [\[3\]](#) we took the average over all grid points and all time instances. The figures show the resulting data for the active vertex model and the multiphase field model, respectively.

Appendix 3

Results using polygonal shape analysis

To visualize the similarities and differences between q_p and γ_p we show the analogue of **Figure 4** [\[4\]](#) using the Minkowski tensors in **Appendix 3 Figure 15** [\[5\]](#) using polygonal shape analysis. For regular polygons, both shape characterization methods behave in the same way as seen in the first row. For irregular shapes, like in the 2nd row, they already show a different behaviour. Note that the rounded shapes from **Figure 4** [\[4\]](#) are missing in **Appendix 3 Figure 15** [\[5\]](#) as γ_p requires a polygon.

The influence of the choice of the shape characterization method is not only visible in the values for single shapes, it can also be seen in the mean and standard deviation. To illustrate this, we use γ_p to characterize the shapes from the same experimental data as in **Figure 1** [\[6\]](#) and **Figure 2** [\[7\]](#). The segmented data consists of smooth contours for each cell. As γ_p only works with vertex coordinates the first step is to identify these. Note that this always leads to the approximation of the cell shape with a polygon and therefore to a simplification of the shape. For finding the vertices we consider the Voronoi interface method (Saye and Sethian, 2011 [\[8\]](#), 2012 [\[9\]](#)). Using the so obtained vertices and [Equation 8](#) [\[10\]](#) and [Equation 9](#) [\[11\]](#) leads to the results shown in **Appendix 3**

Appendix 1—table 1.

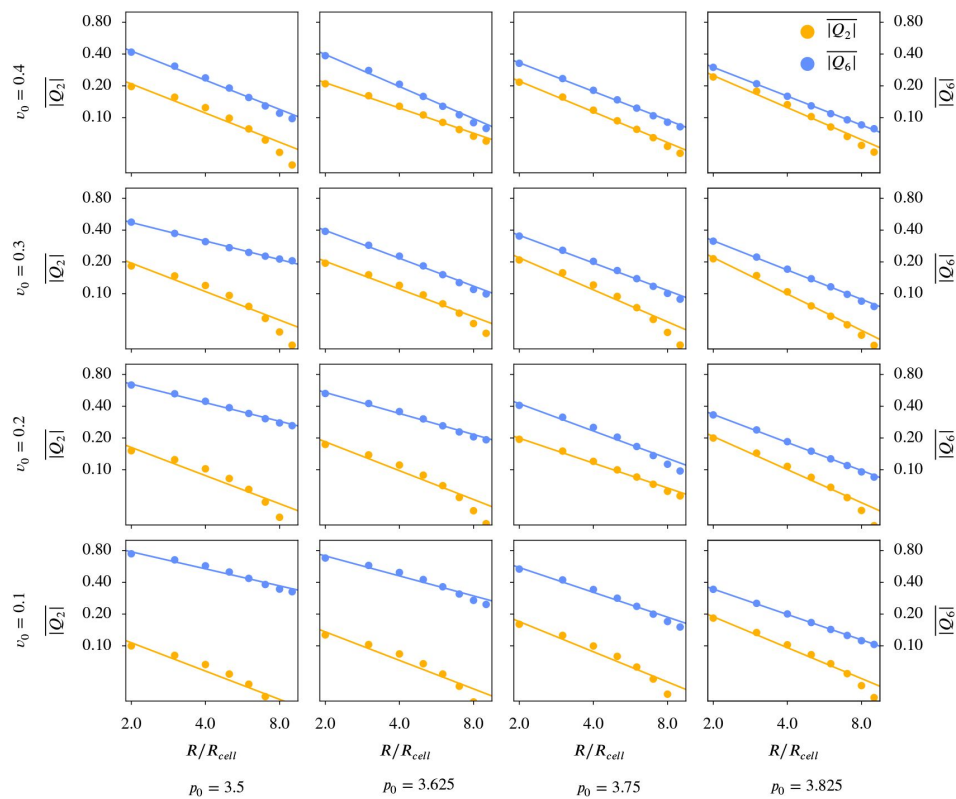
Values of the dimensionless parameters used in the active vertex model

Parameter	Description	Numerical Value
N	Number of cells	100
L	Simulation box size	10
T	Total simulation time	300
k_p	Perimeter elastic modulus	1.0
τ	Simulation time step	0.01
v_0	Self-propulsion strength (i.e. activity)	0.1 - 0.4
D_r	Rotation diffusion coefficient	0.05
p_0	Shape index of active cells	3.5 - 3.875

Appendix 1—table 2.

Values of the dimensionless parameters used in the multiphase-field model simulations

Parameter	Description	Numerical Value
N	Number of cells	100
L	Simulation box size	100
T	Total simulation time	150
ϵ	Interface width	0.15
τ	Simulation time step	0.005
v_0	Self-propulsion strength (i.e. activity)	0.4 - 1.0
D_r	Rotation diffusion coefficient	0.01
α	Alignment parameter	0.1
Ca	Capillary number	0.05 - 0.2
In	Interaction number	0.1
a_a	Cell-cell attraction strength	1.0
a_r	Cell-cell repulsion strength	1.0
h	Mesh size	$5h \approx \epsilon$

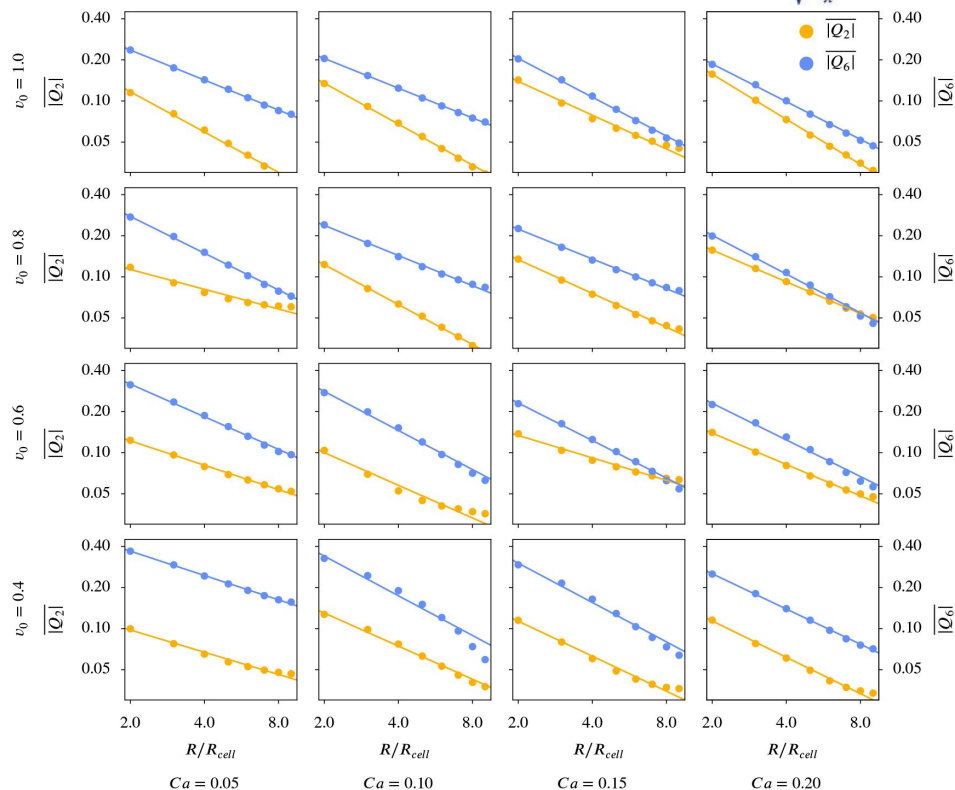


Appendix 2—figure 13.

Q_6 versus Q_2 for different coarse graining radii in the active vertex model, calculated according to [Equation 18](#). A logarithmic scaling was used for both axis. q_p and ϑ_p follow from [Equation 7](#), R_{φ} is chosen as $\sqrt{\frac{A_C}{\pi}}$.

Appendix 2—figure 14.

Q_6 versus Q_2 for different coarse graining radii in the multiphase field model, calculated according to Equation 18. A logarithmic scaling was used for both axis. q_p and ϑ_p follow from Equation 7, R is chosen as $\sqrt{\frac{A_c}{\pi}}$.



Appendix 3—figure 15.

Regular and irregular shapes, adapted from Armengol-Collado et al. (2023), with magnitude and orientation calculated by Equation 8 and Equation 9. The brightness scales with the magnitude $|y_p|$.

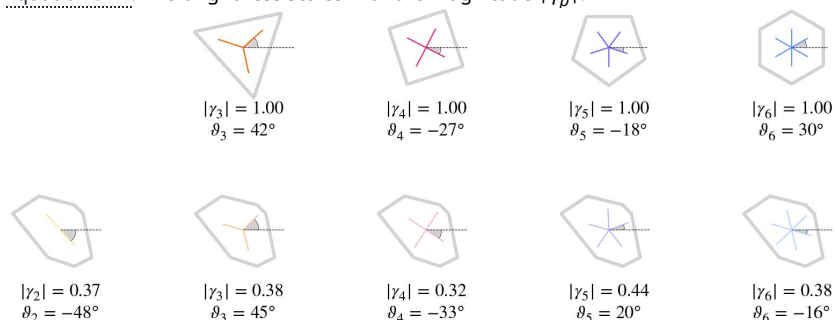


Figure 16 and **Appendix 3 Figure 17**. Compared to **Figure 1** and **Figure 2**, where the Minkowski tensor was used for the same data, we see that the choice of the shape characterization method highly influences the results.

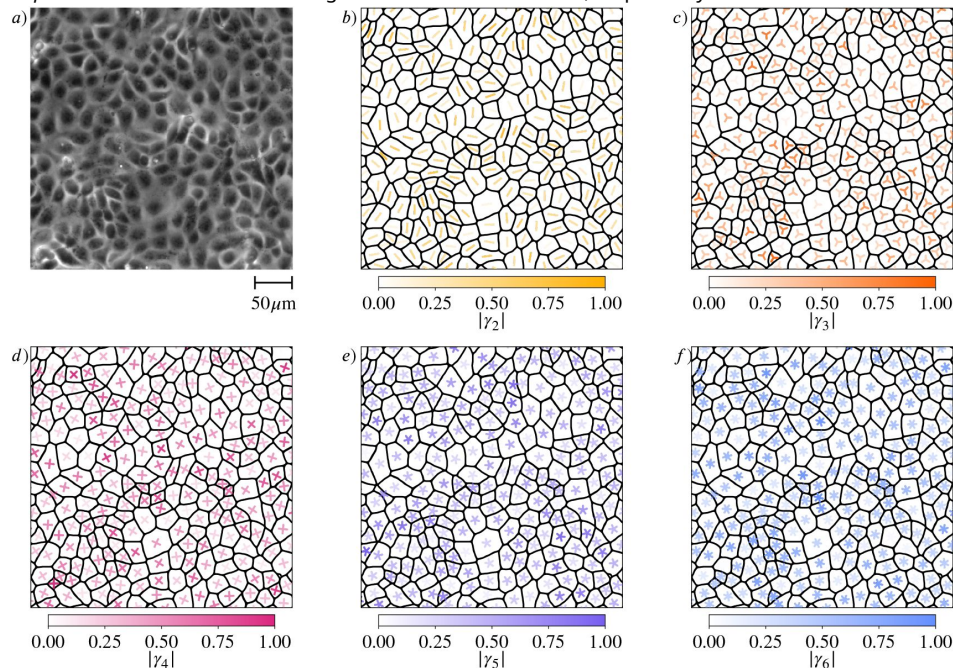
For comparison with the literature we are considering a coarse-graining, following the strategy introduced in Armengol-Collado et al. (2023). We therefore regard the coarse-grained shape function $\Gamma_p = \Gamma_p(\mathbf{x})$, which is the average of all shape functions γ_p of cells whose center of mass $\mathbf{x}_c$ lies within a circle with radius R and center $\mathbf{x}$. In a formula this is:

$$\Gamma_p(\mathbf{x}) = \frac{\sum_{j=1}^{100} \gamma_p(C_j) \Theta(R - |\mathbf{x} - \mathbf{x}_c(C_j)|)}{\sum_{j=1}^{100} \Theta(R - |\mathbf{x} - \mathbf{x}_c(C_j)|)} \quad (19)$$

whereby the formula uses that our simulations contain 100 cells. C_i denotes the i -th cell and Θ denotes the Heaviside step function with $\Theta(x) = 1$ for $x > 0$ and $\Theta(x) = 0$ otherwise. As in Armengol-Collado et al. (2023) the position $\mathbf{x}$ is sampled over a square grid with spacing $R_c = \sqrt{\frac{A_c}{\pi}}$ with A_c the cell area. For **Appendix 3 Figure 20** we took the average of Γ_p over all grid points and all time instances. The figure shows the resulting data for the active vertex model.

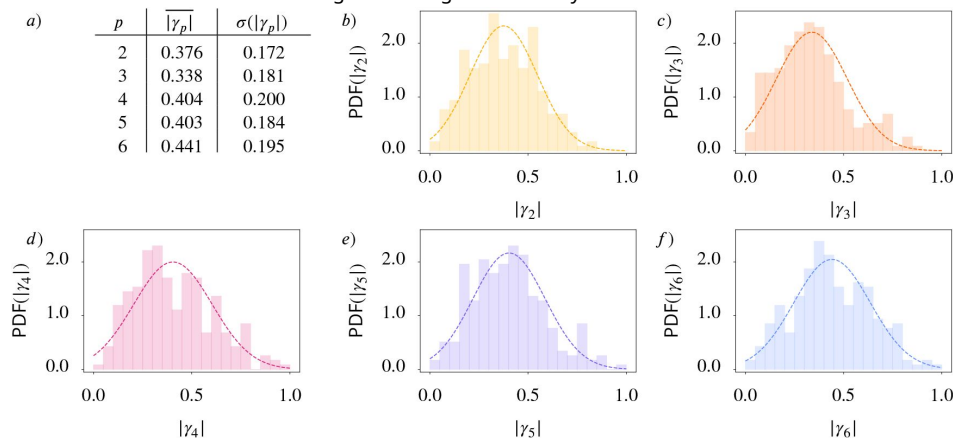
Appendix 3—figure 16.

Shape classification of cells in wild-type MDCK cell monolayer. a) Raw experimental data. b) - f) Polygonal shape classification, visualized using γ_p calculated by Equation 8 and Equation 9 for $p = 2, 3, 4, 5, 6$, respectively. The brightness and the rotation of the p -atic director indicates the magnitude and the orientation, respectively.



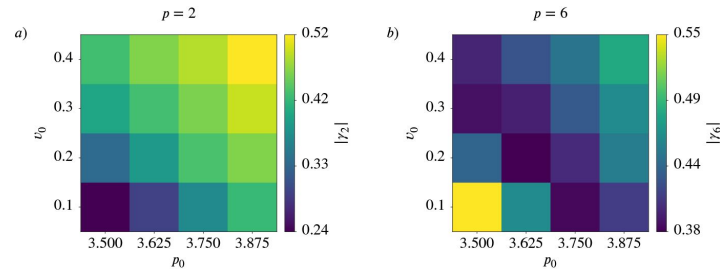
Appendix 3—figure 17.

Statistical data for cell shapes identified in Appendix 3 Figure 16. a) Mean $\overline{|\gamma_p|}$ and standard deviation $\sigma(|\gamma_p|)$ of $|\gamma_p|$. b) - f) Probability distribution function (PDF) of $|\gamma_p|$ for $p = 2, 3, 4, 5, 6$, respectively. Normal distributions with the same mean and standard deviations are added to the histograms as a guide to the eye.



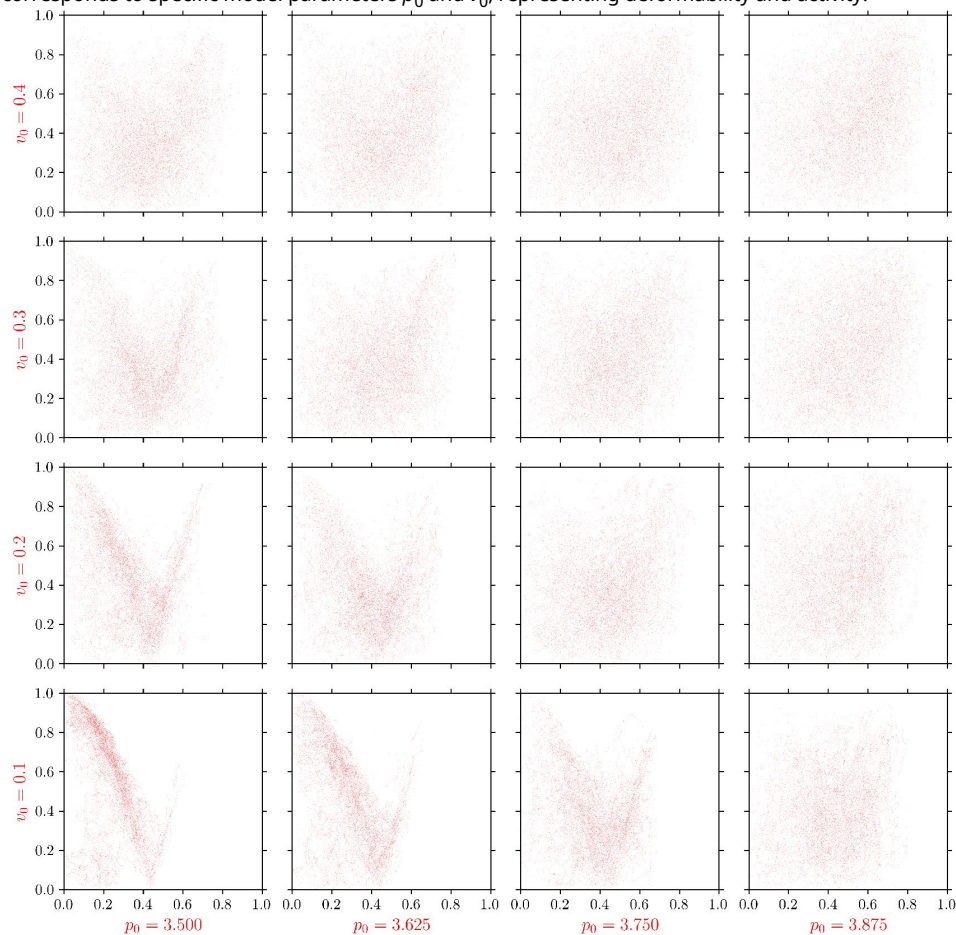
Appendix 3—figure 18.

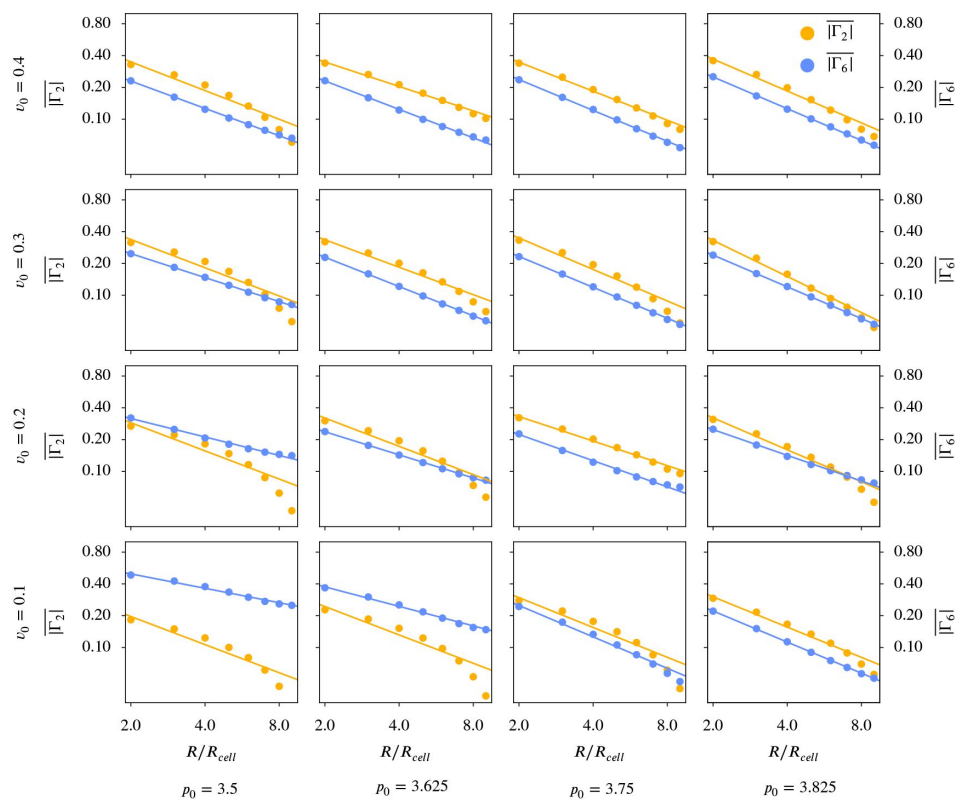
Mean value $\overline{\gamma_p}$ as function of deformability p_0 and activity v_0 for active vertex model. a) nematic order ($p = 2$), b) hexatic order ($p = 6$). While the behaviour of $|\gamma_2|$ qualitatively agrees with the behaviour of q_2 , the behaviour of $|\gamma_6|$ is quite different to the one of q_6 .



Appendix 3—figure 19.

y_6 (y-axis) versus y_2 (x-axis) for all cells in the active vertex model. For each cell and each timestep we plot one point (y_2, y_6). Each panel corresponds to specific model parameters p_0 and v_0 , representing deformability and activity.





Appendix 3—figure 20.

Γ_6 versus Γ_2 for different coarse graining radii in the active vertex model, calculated according to [Equation 19](#). A logarithmic scaling was used for both axis. γ_p follows from [Equation 8](#), R_ℓ is chosen as $\sqrt{\frac{A_c}{\pi}}$.

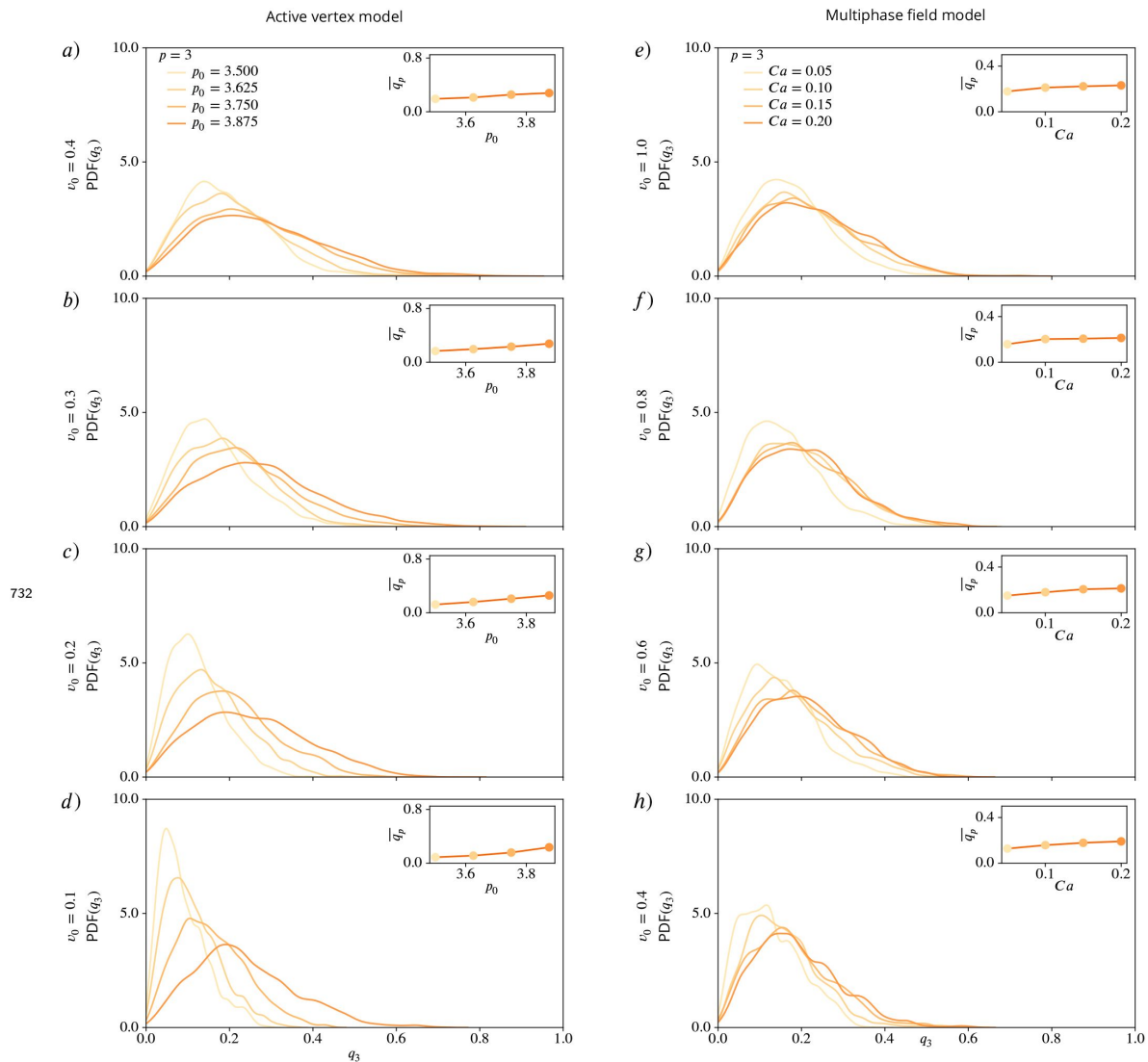


Figure 8—figure supplement 1.

PDFs for q_3 using kde-plots, for varying deformability p_0 or Ca and fixed activity v_0 . Inlets show mean values of q_3 as function of deformability. a) - d) active vertex model, e) - h) multiphase field model for decreasing activity.

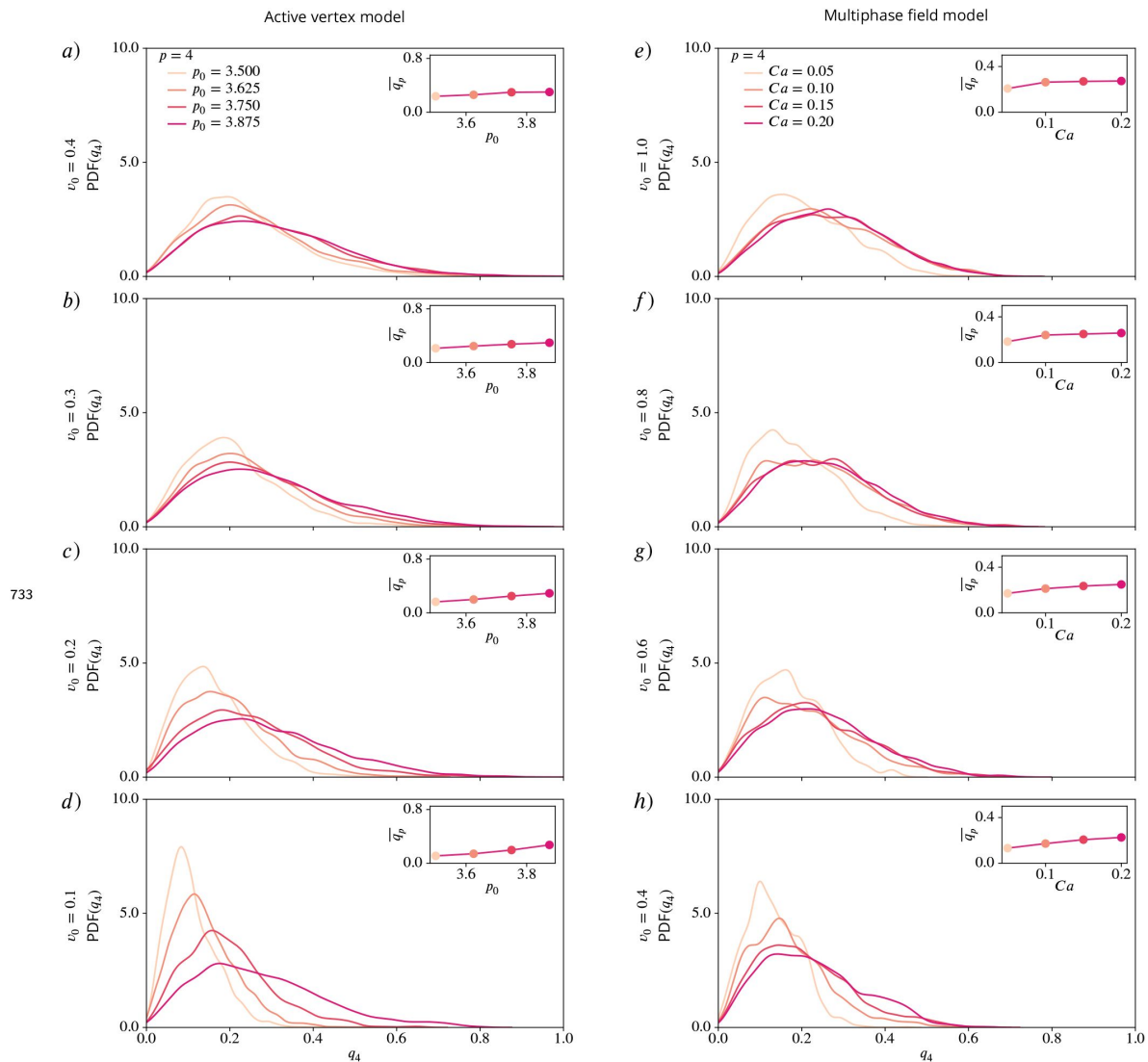


Figure 8—figure supplement 2.

PDFs for q_4 using kde-plots, for varying deformability p_0 or Ca and fixed activity v_0 . Insets show mean values of q_4 as function of deformability. a) - d) active vertex model, e) - h) multiphase field model for decreasing activity.

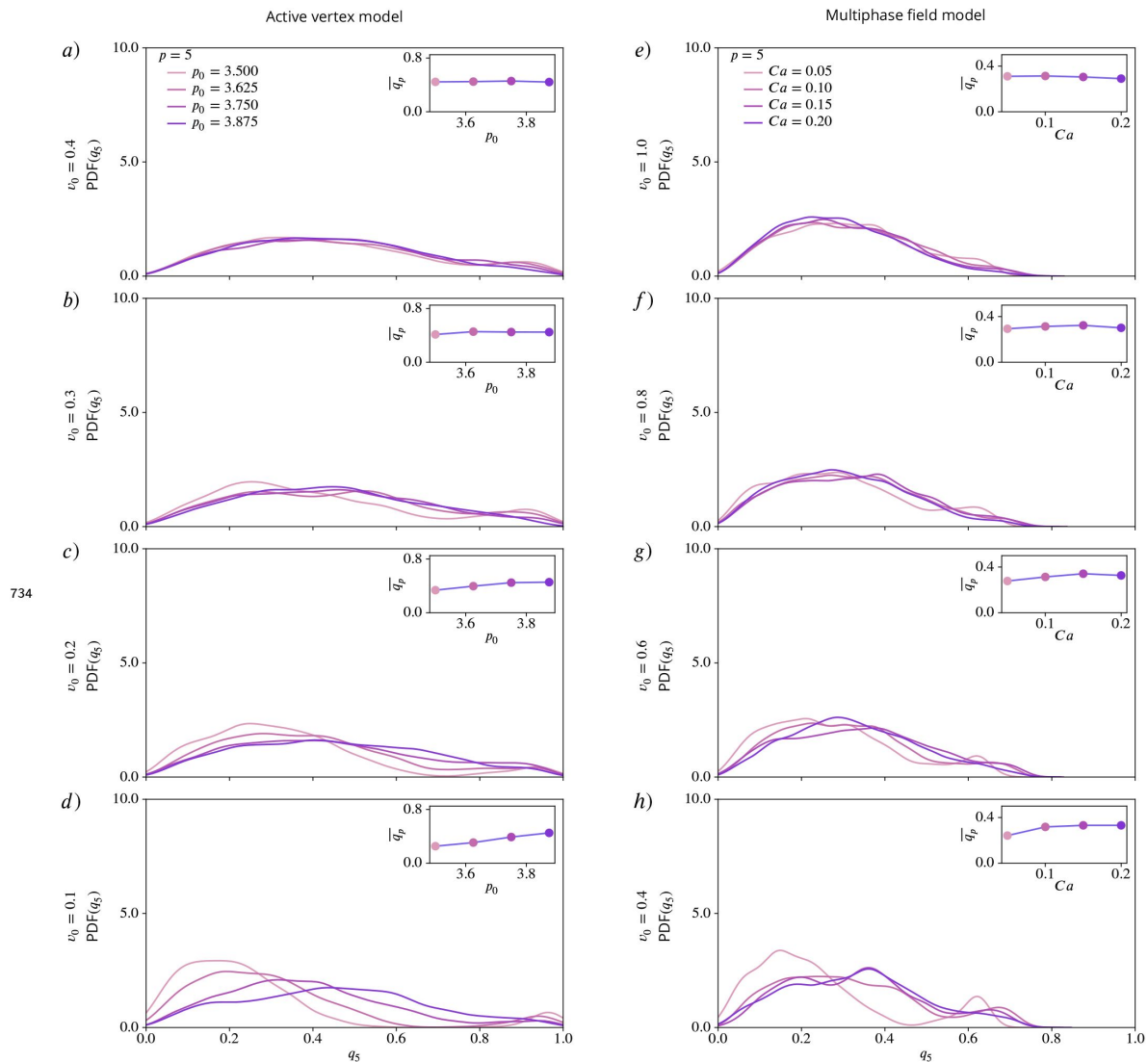


Figure 8—figure supplement 3.

PDFs for q_5 using kde-plots, for varying deformability p_0 or Ca and fixed activity v_0 . Insets show mean values of q_5 as function of deformability. a) - d) active vertex model, e) - h) multiphase field model for decreasing activity.

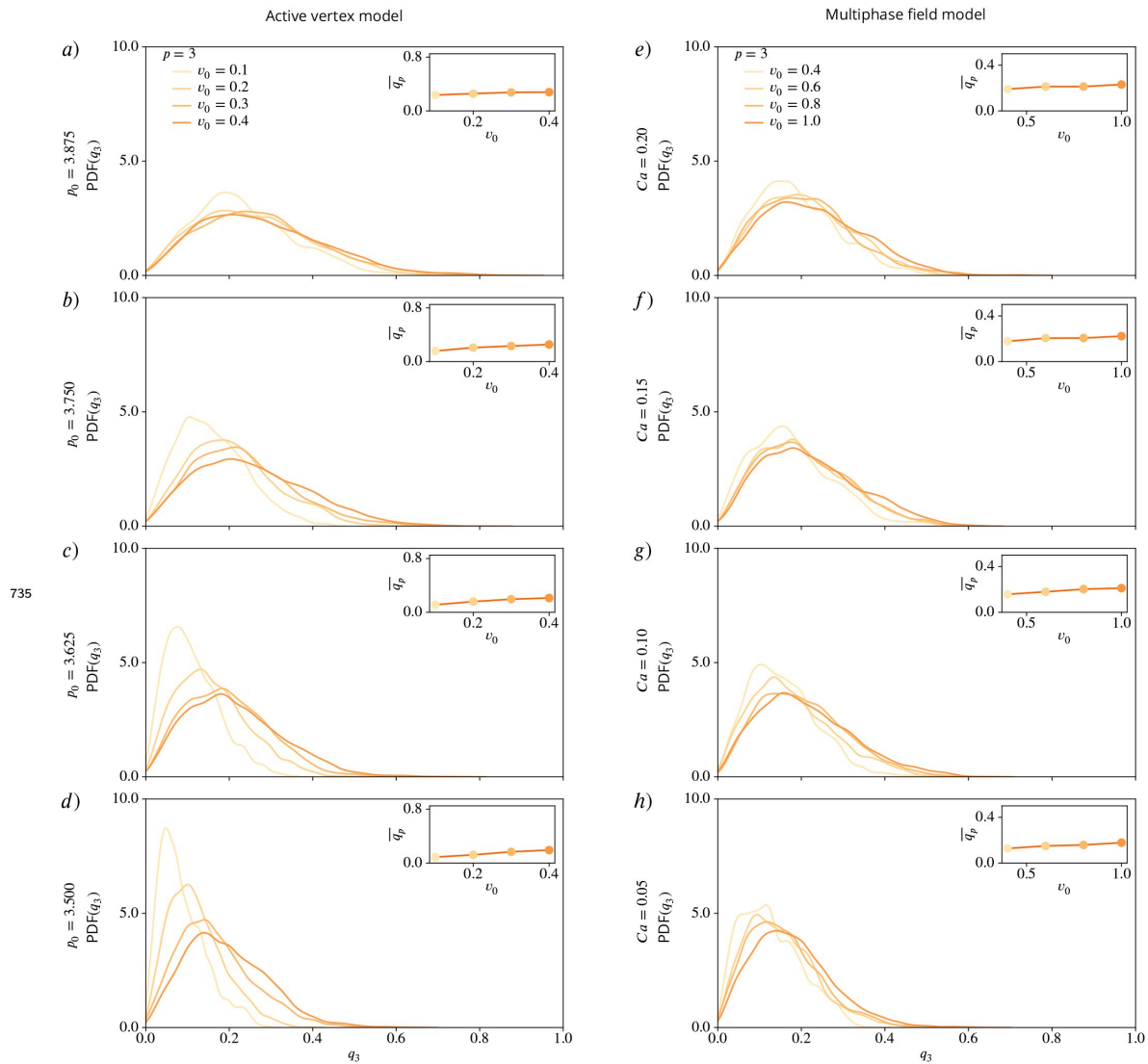


Figure 9—figure supplement 1.

PDFs for q_3 using kde-plots, for varying activity v_0 and fixed deformability p_0 or Ca . Insets show mean values of q_3 as function of activity. a) - d) active vertex model and e) - h) multiphase field model for decreasing deformability.

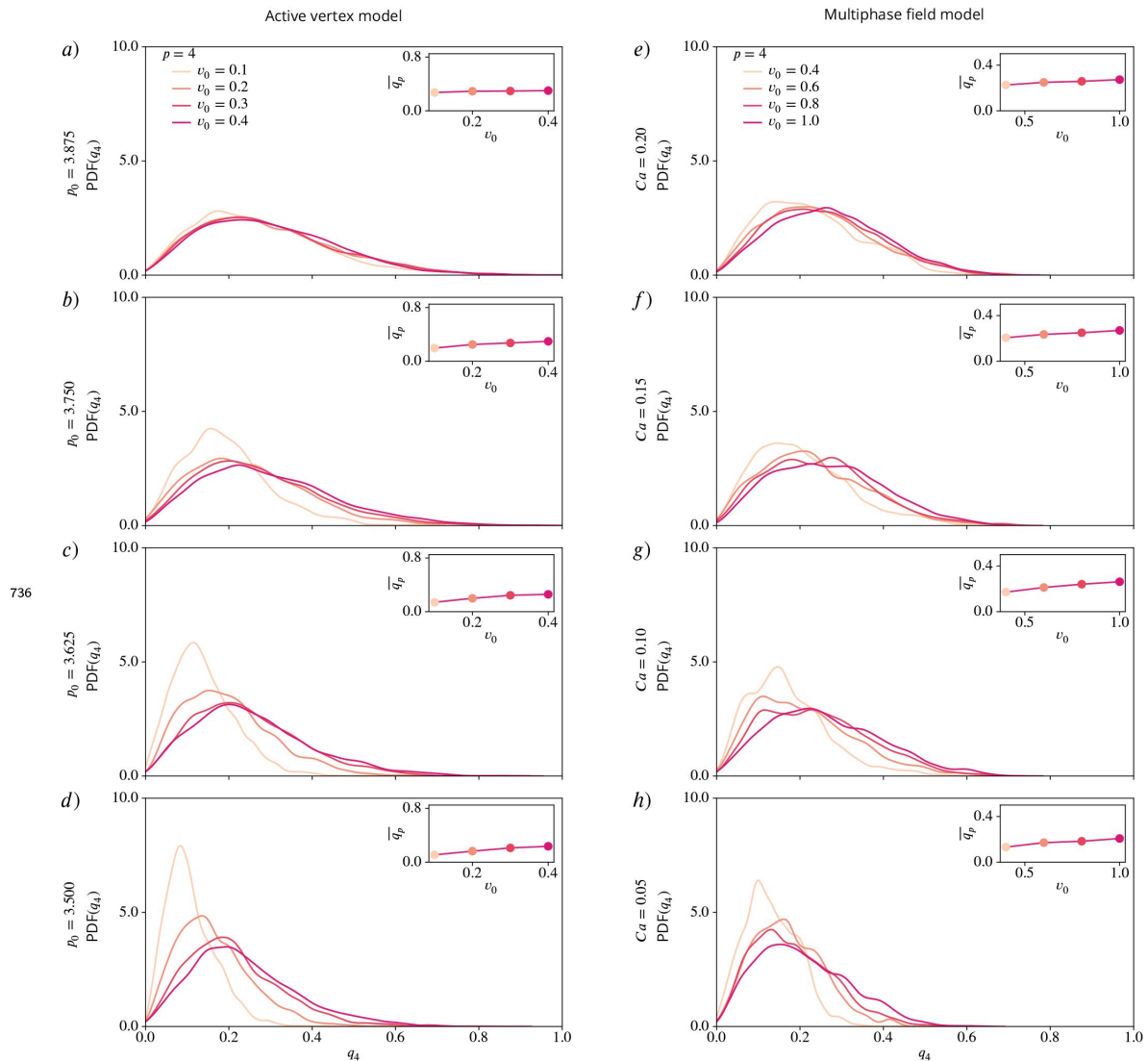


Figure 9—figure supplement 2.

PDFs for q_4 using kde-plots, for varying activity v_0 and fixed deformability p_0 or Ca . Insets show mean values of q_4 as function of activity. a) - d) active vertex model and e) - h) multiphase field model for decreasing deformability.

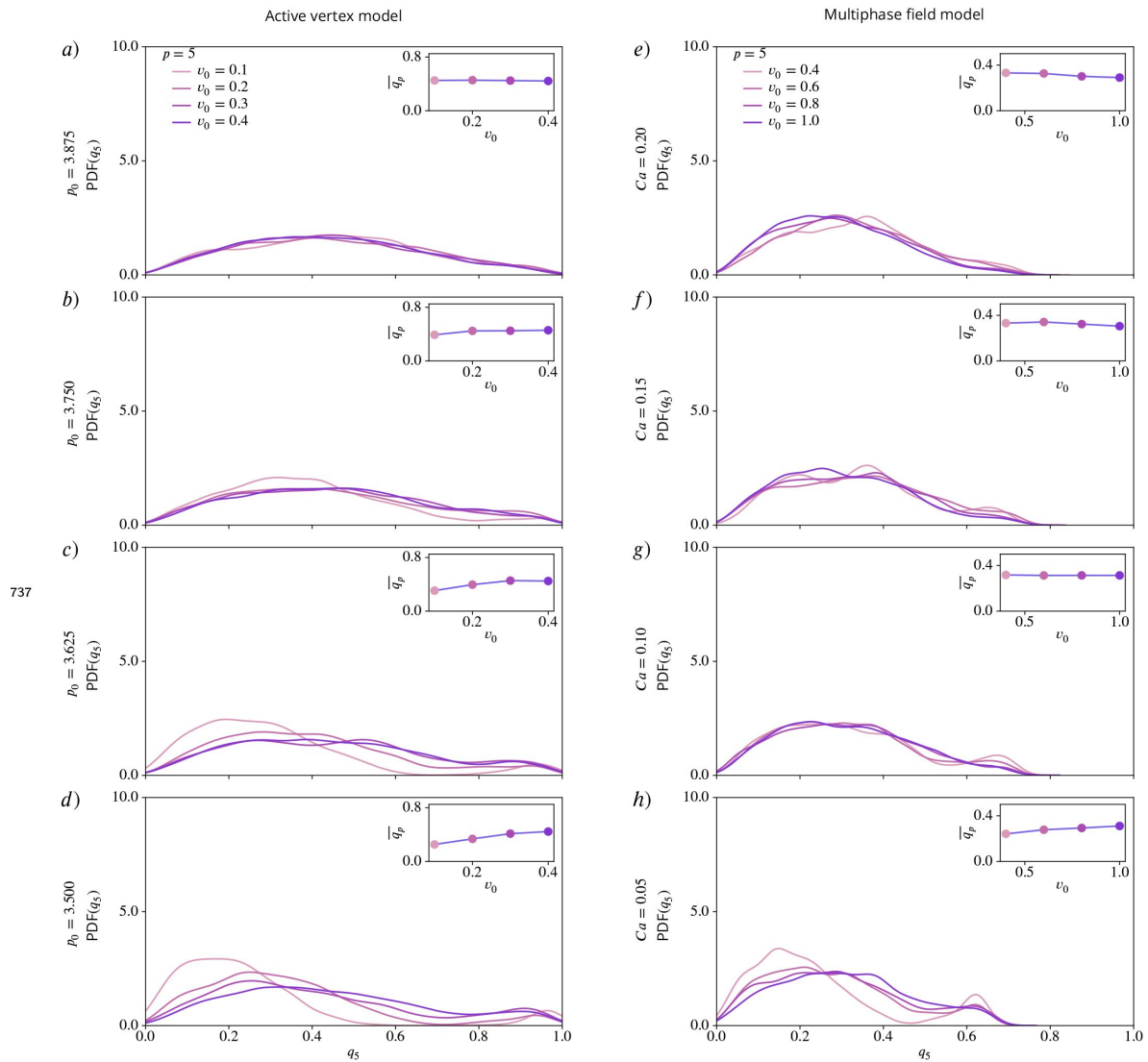


Figure 9—figure supplement 3.

PDFs for q_5 using kde-plots, for varying activity v_0 and fixed deformability p_0 or Ca . Insets show mean values of q_5 as function of activity. a) - d) active vertex model and e) - h) multiphase field model for decreasing deformability.

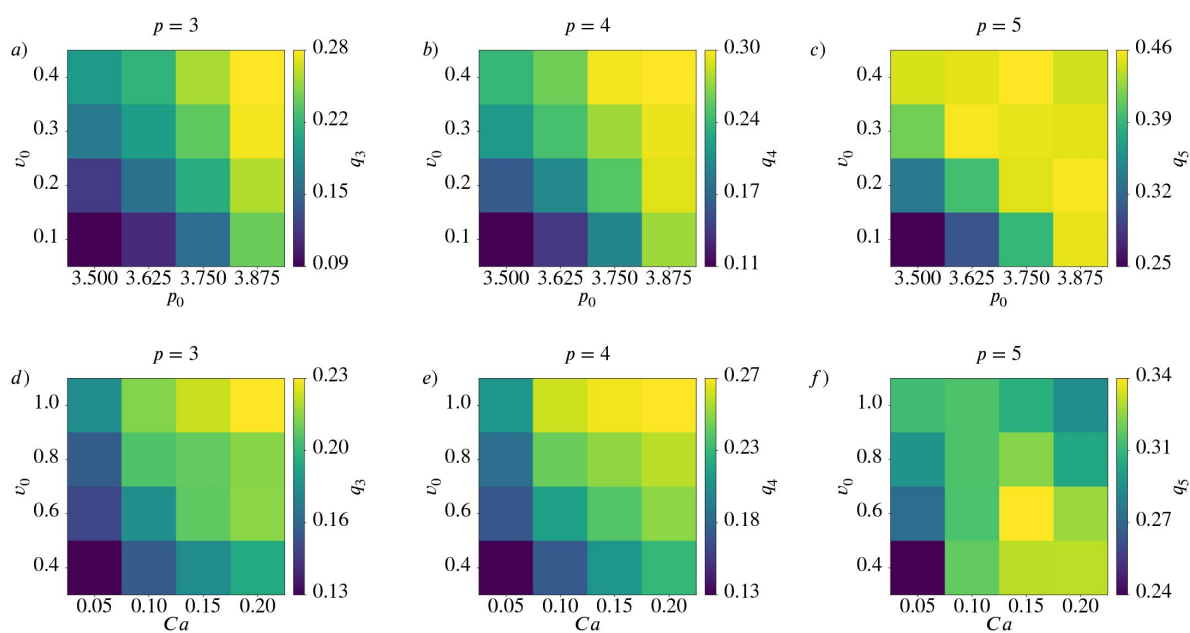


Figure 10—figure supplement 1.

Mean value $\overline{q_p}$ for $p = 3$ (left), $p = 4$ (middle) and $p = 5$ (right) as function of deformability p_0 or Ca and activity v_0 for active vertex model (a - c) and multiphase field model (d - f).

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Reviewer #1 (Public review):

Summary:

The authors' stated aim is to introduce so-called Minkowski tensors to characterize and quantify the shape of cells in tissues. The authors introduce Minkowski tensors and then define the p-atic order q_p , where p is an integer, as a cell shape measure. They also introduce a previously defined measure of p-atic order in the form of the parameter γ_p . The authors compute q_p for data obtained by simulating an active vertex model and a multiphase field model, where they focus on p=2 and p=6 - nematic and hexatic order - as the two values of highest biological relevance. Based on their analysis, the authors claim that q_2 and q_6 are independent, that there is no crossover for the coarse-grained quantities, that the comparison of q_p for different values of p is not meaningful, and determine the dependence of the mean value of q_2 and q_6 on cell activity and deformability. They then apply their method to data from MDCK monolayers and argue that the γ_p "fail to capture the nuances of irregular cell shapes".

Strength:

The work presents a set of parameters that are useful for analyzing cell shape.

Weaknesses:

The main weakness of the manuscript is that the points that the authors make are not sufficiently elaborated or supported by the data. Although they start out with Minkowski tensors, they eventually only consider the parameters q_p , which can be defined without any recourse to Minkowski tensors. Also, I dare to doubt that the average reader will benefit from the introduction to Minkowski tensors as it remains abstract and does not really go beyond repeating definitions. Eventually, for me, the work boils down to the statement that when you want to characterize (2d) cell shape, then it is better to take the whole cell contour instead of only the positions of the vertices of a polygon that approximates the full cell shape. By the way, for polygons, the q_p and γ_p should convey the same information as the vertex positions contain the whole geometric information.

Some statements made about the values of q_p are not supported by the data. For example, an independence of values of q_2 and q_6 cannot be inferred from Figure 7. Actually, Figure 8 points to some dependence between these values as the peaks of the pdfs move in the opposite direction as deformability and activity are changed. Figure 1 suggests that in general, larger cells have lower values of q_p for all p. Some more serious quantification should be obtained here.

The presented experimental data on MDCK cells is anecdotal.

<https://doi.org/10.7554/eLife.105680.1.sa3>

Reviewer #2 (Public review):

Summary:

Oriental symmetries of cells and tissues play an important role in describing processes in development and disease, and the methods used to investigate them rely on the detection of cell shape. In this interesting and very timely manuscript by Lea Happel et al., Minkowski tensors are introduced to study the orientational symmetries of cells and set in comparison to existing shape descriptors, such as the shape function introduced by Armengol-Collado et al., which captures the orientational symmetry by the vertex positions of the polygonal shape of the cell. As an advantage, the Minkowski tensors consider the real cell shape with its

arbitrary curvature of the cortex. Using computational models, such as the active vertex model and the multiphase field model, as well as experimental support with MDCK monolayers, the authors find that the orientational symmetries are independent of one another, as well as that they are dependent on the activity and deformability of the cells, resulting in a monotonic trend. A trend that has not been observed for the hexatic symmetry using the shape function. Together with the lack of hexatic-nematic crossover at the tissue scale, the authors suggest a reconsideration of findings from other shape descriptors. Taken together, the Minkowski tensors set a framework to investigate orientational symmetries at a single cell scale and how they may interplay in biological tissues.

Strengths:

The authors introduce the Minkowski tensors, which capture the p-atic orders of cells in tissues, considering their real shape instead of a polygonal approximation as reported for other shape descriptors in the literature. Thus, they do not depend on the vertex positions of the cells nor on the number of neighboring cells. The Minkowski tensors capture the dependence of the p-atic orders on the cell activity and deformability in a monotonic manner, which makes them a robust tool for quantifying p-atic orders at a single-cell scale, especially for rounded cells. The robustness has been tested by comparing the results of two computational model systems that simulate cell monolayers and whose results have been extended with experimental data. The Minkowski tensors have been used to explore the role of cell-cell adhesion and density in epithelial cells and have shown similar results to the shape function, a polygonal shape descriptor.

Weaknesses:

The authors point out the importance of studying the orientational order in biological systems. However, the current version of the manuscript lacks statistical information, a description of analysis methods, and experimental support. This support is needed to strengthen (i) the results of the two computational models and (ii) give weight to the authors' strong claim against other widely accepted shape descriptors capturing p-atic orders. The Minkowski tensors, which consider the real cell shapes, are reported to be a better method to investigate the p-atic orders of cells than the shape function introduced by Armengol-Collado et al. While there may be differences in the reported results coming from the two different approaches, both approaches show similar trends. As it stands, there is substantiated discussion as to why one method would be better than the other. The shape function, γ_6 , may not be monotonic for great changes in cell activity and deformability, hinting at a potential weakness. In contrast to the shape function and results by Armengol-Collado et al. and Eckert et al., the coarse-grained Minkowski tensors do not capture the hexatic-nematic crossover at the tissue scale, applied here only to computational models. The cells simulated in the computational models have a similar size and the monolayer has a nearly regular pattern, which does not reflect the density variance in biological tissues. To strengthen the author's claim that there is no crossover at the tissue scale, experimental verification is essential. Further, the robustness of the Minkowski tensors seems to rely on determining the p-atic orders on the shape of individual cells in the tissue. However, when applying the shape descriptor to experimental systems, the p-atic orders are very low, perhaps too low for comparisons between different p-atic orders with meaningful conclusions.

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Reviewer #3 (Public review):

Hapel et al. submit an article entitled “Quantifying the shape of cells - from Minkowski tensors to p-atic order”. The paper reports the p-atic quantitative method - established in physics - to extract cell shapes in experiments using phase contrast images of MDCK cells and

simulations - vertex model and phase fields. The rationale of the quantification with adaptation of Minkowski tensors, as well as the detailed extraction of distributions of shapes and plots, distributions quantifying shapes are documented, with an emphasis on changes in cell shapes and their importance in epithelial dynamics.

Higher rank tensors are considered as well as representations with intuitive meanings and q_i orders and their potential correlations or absence of correlations. For example, q_2 and q_6 , and statements about nematic and hexatic orders. A strong body of evidence is already reported in the papers of Armengol et al., quoted substantially in the paper, and the authors insist on an improvement thanks to the Minkowski tensors approach to challenge the former crossovers correlations statements.

Although the approach seems to present advantages, the paper does not appear sufficiently novel. Beyond the Armengol et al. paper, the advantages of this approach compared to the shear decomposition (from MPI-PSK Dresden) or the links joining centroids and its neighbours approach (MSC/Curie Paris) for example.

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Author response:

We thank the editors and the reviewers for their valuable comments. In response to these suggestions, we will add rigorous statistical measures and extend the experimental support of our findings in a revised version. Indeed, as we will show, doing so strengthens all the main claims. Specifically:

Concerning Reviewer 1:

- It is important to emphasise that the advantage of deriving shape measures q_p from Minkowski tensors is their robustness and stability, that is well-established from extensive, rigorous mathematical analyses. Introducing q_p without this connection to revised Minkowski tensors would not allow to claim this stability property for the considered measures.
- Even though for a polygon the vertex positions contain the whole geometric information, using q_p and γ_p lead to different results, see Fig. 6 for an example.
- We wholeheartedly agree that our statement on independence of values of q_2 and q_6 can be extended and more quantitatively established by rigorous statistical measures. This is exactly what we will do in the revised version, not only providing statistical measures on the presented data, but also extending our analyses to the published data from Armengol-Collado JM, Carenza LN, Eckert J, Krommydas D, Giomi L. Epithelia are multiscale active liquid crystals. *Nature Physics*. 2023; 19:1773–1779. As we shall show these analyses further strengthen this claim, unequivocally establishing the independence of q_2 and q_6 in two different models (active vertex model and multiphase-field model), as well as two different sets of experiments (the ones in the original manuscript, and the published one from Armengol-Collado JM, Carenza LN, Eckert J, Krommydas D, Giomi L. Epithelia are multiscale active liquid crystals. *Nature Physics*. 2023; 19:1773–1779).

Concerning Reviewer 2:

To fully address this point, we have extended our analyses to explore the published data of Armengol-Collado JM, Carenza LN, Eckert J, Krommydas D, Giomi L. Epithelia are multiscale active liquid crystals. *Nature Physics*. 2023; 19:1773–1779. As we shall show in the revised manuscript, the crossover between nematic and hexatic is only specific to the use of γ_p for characterizing the shape and coarse-graining of the associated order. Using q_p as the shape

measure this crossover disappears. Therefore, this analyses concretely demonstrate that the crossover is not a robust physical feature of the system and is dependent on the method used to define shape characteristics.

Concerning Reviewer 3:

We respectfully note a misunderstanding from the referee: The briefly mentioned approaches of other groups, turn out to be not measuring shape but connections between cells. Conceptually these approaches are therefore related to bond order parameters. We already comment at the end of the section introducing Minkowski tensors that bond order parameters cannot quantify the shape of a cell. The same argumentation also holds for other such approaches. In our revised version we will further clarify this distinction, to avoid any confusion or misinterpretation.

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